On the Impossibility of Lossless Convexification for Attitude-Rate-Limited (5-DOF and 6-DOF) Powered Descent, and a Sound Single Convex Program for Gimballed Landers

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11 November 2026

Abstract

Lossless convexification (LCvx) solves the three-degree-of-freedom (3-DOF) powered-descent guidance problem, including the nonconvex minimum-thrust constraint, by one convex program. Its relaxed solutions are provably feasible for the original problem. We ask whether the same is possible once the thrust direction is tied to a vehicle attitude that turns at a finite rate. This is the case for every thrust-vector-controlled (TVC) lander. We give a negative answer for the class of relaxations that LCvx belongs to: convex programs in which the physical state is a decision variable and the boundary data enter affinely. No such relaxation can be lossless for 5-DOF (translation plus thrust-axis pointing) or kinematic (attitude-rate-limited) 6-DOF descent. Our headline result, machine-checked in Lean 4/Mathlib, covers the state-constrained problem (with constant thrust-acceleration bounds, i.e. constant mass): with altitude and glide-slope constraints, for every final time, with free or upright terminal attitude, in 5-DOF and in kinematic 6-DOF, no convex relaxation of this class is lossless, whatever its lifted variables and cost. The same constrained impossibility is also machine-checked with explicit mass depletion, with free final time on compact windows, and at the motivating vehicle's actual constants (kinematic attitude model, 15° cone). The obstruction is geometric. A lossless relaxation of this kind forces the set of feasible boundary conditions to be convex (Theorem A). For attitude-rate-limited models this set is not convex. The reason is that the vehicle cannot chatter its thrust direction. We prove this without state constraints, with altitude and glide-slope constraints for every horizon, for upright landings, for free final time on compact windows, for line search over the final time, without a minimum throttle (without state constraints), and for kinematic and independent-torque 6-DOF models. A per-instance version shows that every state-linear relaxation of a minimum-landing-error problem is lossy on explicit feasible instances, and an optimization version shows that no uniform relaxation is value-lossless for all convex costs. On the positive side, LCvx is exact once the attitude profile is fixed. For the gimbal-torque-coupled TVC rigid body with constant mass, the problem with a given attitude profile is exactly convex, with no relaxation needed. So all of the nonconvexity lies in the choice of attitude profile. Degree-2 semidefinite lifts are not exact in general, but they give a posteriori global-optimality and infeasibility certificates, and they are certified exact on a coplanar subclass. Besides the headline theorem, the unconstrained obstruction (with mass depletion), the abstract per-instance, line-search and value-losslessness results, and an explicit hard-landing instance of the per-instance obstruction are machine-checked. For the real pure-TVC actuation, landing upright at rest with altitude and glide-slope constraints, nonconvexity is proved by computer-assisted certificates checked in exact arithmetic (horizons of 1 and 2 s without a tilt cap, and 3 and 5 s in the tilt-capped class; constant mass, unlimited gimbal rate). Numerically, the obstruction is thin for in-air descents of a small lander (centimetre scale), but reaches metre scale for pad-launched hops. In simulation, the vehicle's 3-DOF guidance claims landings no flyable plan reaches, while planners that iterate on the attitude (standard successive convexification, or one built on the exact convex inner problem) reach the 6-DOF

optimum and land reliably. Finally, the impossibility concerns programs that are both sound and complete. Giving up completeness, we plan the path of the vehicle's centre of percussion, in whose coordinates the gimballed rigid body is differentially flat, as one second-order cone program with the actuator limits imposed as convex restrictions. We prove that every feasible point is a trajectory of the pure-TVC model, with depleting mass, that meets the throttle, gimbal and tilt limits at every instant (not machine-checked). In simulation this single program costs 0.1% of fuel at the operating point, never returned an unflyable plan, and, replanning ten times a second, landed within 1.1 cm in every in-air case, against 18 cm for successive convexification once each planner's computation time is counted.

How to read and check this paper

Status tags. Each result carries a tag saying how it is established: [Lean-checked] machine-checked in Lean 4/Mathlib (no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound` only); [proved; not machine-checked] pen-and-paper proof; [proved in Appendix E; not independently refereed, not machine-checked] full proof given here, checked numerically but by no second reader (Theorem 4.7 only); [proved, computer-assisted] proof by certificate checked in exact (rational and ball) arithmetic; [proof sketch only]; [numerical evidence only]; [open].

Checking the formal results. The supplementary archive contains the Lean project. After `lake exe cache get` and `lake build`, the command `lake env lean Verify.lean` prints the headline statements and confirms that every Lean declaration cited here exists and uses only the three standard axioms (Section 7). The archive also contains the scripts and data of the computer-assisted certificates of Section 3.8.

Statement on methods. This work was carried out with extensive use of an AI system (Claude, Anthropic), which drafted the proofs, the Lean formalization, the numerical experiments and the text under the authors' direction. The Lean-checked results do not depend on trusting that process. The pen-and-paper proofs and the computer-assisted certificates were cross-checked by independent verification passes, with the exception of Theorem 4.7, which was added later and is tagged accordingly. This paper has not yet been peer reviewed.

1 Introduction

Lossless convexification. Powered-descent guidance asks for a thrust history that takes a lander from its current position and velocity to rest at a target, subject to thrust-magnitude bounds $0 < \rho_1 \le \|T\| \le \rho_2$, pointing and state constraints, and often minimum fuel or minimum landing error. The lower thrust bound makes the control set an annulus, which is nonconvex. The lossless convexification (LCvx) programme of Açıkmeşe and coauthors [1–5] replaces the annulus by a convex cone in an augmented space, $\|T\| \le \Gamma$, $\rho_1 \le \Gamma \le \rho_2$, and proves that optimal solutions of the relaxed second-order cone program (SOCP) satisfy $\|T\| = \Gamma$ almost everywhere. They are therefore optimal for the original problem. Extensions cover pointing constraints [4, 6], state constraints [7, 8], some nonlinear dynamics [9], normality-based general theory [10], annular control sets under controllability [11] and discrete time [12]. Surveys and tutorials are [13, 14]. LCvx-based guidance has flown [15, 16]. Every one of these results treats the thrust direction as a free control variable, possibly restricted to a cone about a fixed or prescribed axis. Six-degree-of-freedom (6-DOF) powered-descent methods use sequential convex programming instead [17–21], and LCvx appears there only as an embedded relaxation of the thrust bounds [14]. A search of the literature through November 2026 found no LCvx theorem for a model in which the thrust direction is a rate-limited state, and no impossibility result for lossless convexification beyond the discrete-time counterexamples of [12]. A separate line of work plans in the flat outputs of a differentially flat system [22–25], with splines whose control points enforce constraints between nodes [26]; we use it in Section 4.5.

Why 5-DOF and 6-DOF matter. On a gimballed (TVC) lander the thrust acts along an engine axis that can deflect by at most a gimbal angle δ from the body axis b. The body axis itself turns at a finite rate, and on a pure-TVC vehicle it is turned *by* the gimbal. A 3-DOF LCvx solution can ask for thrust-direction changes the vehicle cannot follow. The lander that motivated this study is a small throttleable TVC vehicle ($\approx$ 80 kg wet, thrust 400–1000 N, $\pm 15^\circ$ box gimbal, usable body rate 0.3–0.5 rad/s). Its stored 3-DOF LCvx solutions contain thrust-direction jumps of 15° within one 24 ms control step, plus 180° azimuth flips, which would require angular accelerations around 10^3 rad/s^2. The vehicle can deliver 1–3.5 rad/s^2. The question we address is whether *any* LCvx-type convex program can be lossless once the attitude dynamics are part of the model.

The answer in one paragraph. 3-DOF LCvx works because the reachable set of a linear system with a measurable control is convex: whatever the relaxed problem does on average can be realized by fast switching (Lyapunov–Aumann [27, 28]). In 5-DOF the thrust direction is tied to a body axis that rotates at a finite rate, so it cannot switch. Take a maximal tilt to the left and a maximal tilt to the right. The average of these two manoeuvres is not flyable. Any convex program that contains both must also contain their average, provided the physical state is a decision variable and the boundary data enter affinely. That is exactly the LCvx situation. The rest of the paper makes this precise, removes the idealizations one by one (state constraints, minimum throttle, terminal attitude, final time, 6-DOF), quantifies the size of the obstruction, and describes what can still be convexified exactly. It also shows what the impossibility leaves room for. A lossless program is sound and complete at once. A convex program that is only sound, returning nothing but flyable plans and refusing some landings that exist, is not excluded, and for a gimballed lander we construct one (Theorem 4.7).

Headline result. The following statement is machine-checked in Lean 4 with Mathlib (Section 7). Its proof is Theorem A combined with Theorem B* (Section 3.3).

Main Theorem (machine-checked)**.** *Consider 5-DOF powered descent with rate-limited thrust axis* ($\angle(b(s), b(s')) \le \omega|s - s'|$, $b(0) = e_z$)*, gimbal cone* $\angle(d, b) \le \delta$ *with* $0 < \delta < \pi/2$*, thrust acceleration between constants* $0 < a_{\min} \le a_{\max}$ *with* $g_0 < a_{\max}$*, and soft landing at* t_f *with* $r_z \ge 0$ *and a glide slope* $\tan\gamma \, \|r_h\| \le r_z$ *along the whole trajectory (or any state constraint whose admissible set* $\mathcal{X}$ *contains this cone, e.g. altitude only). Then for every* $t_f > 0$ *and every* $\omega > 0$*, no convex set* $C \subseteq \mathbb{R}^6 \times Y$ *(*Y *an arbitrary real vector space) is lossless for the set of feasible initial positions and velocities, in the feasibility form or in the LCvx optimization form for any cost. The same holds with an upright terminal attitude* $b(t_f) = e_z$*, and for kinematic 6-DOF attitude (paths of proper rotations with body rate at most* ω *and free roll angle), with free or upright terminal attitude.*

The constant acceleration bounds make this formal statement exact for constant mass. Lean also proves the following extensions (Table 2):

- *Mass depletion,* for every t_f: thrust $T \in [\rho_1, \rho_2]$, $m = m_0 - \alpha \int T \ge m_{dry}$, $a = T/m$, with $0 < \rho_1 < \rho_2$, $\alpha > 0$, $0 < m_{dry} \le m_0 - \alpha\rho_2 t_f$ and $g_0 < \rho_2/m_0$; also with upright landing and in kinematic 6-DOF.
- *Free final time* on compact windows, both with constant bounds and with mass depletion.
- *A quantitative version* for $\delta \le \pi/4$: the witness midpoint lies outside the closure of the feasible set, at an explicit distance.

- *An instance at the motivating vehicle's constants:* 400/1000 N, $m_0 = 79.88$ kg, $m_{dry} = 55.88$ kg, $I_{sp} = 180$ s, 15° cone, any $\omega > 0$, any $t_f \leq 42$ s, kinematic 6-DOF with upright landing. A 5-DOF free-time version holds on any window in $(0, 42]$ s.

The pen-and-paper statement, with proof, is Theorem 3.7.

Contributions.

1. *Definitions* (Section 2). We define uniform and state-linear convex relaxations with affine (or separable-monotone) data entry, and lossless relaxations in a feasibility (detection-sound) form and in an optimization form. We show that the standard LCvx definition with attainment implies the feasibility form. We also state precisely what the class excludes: occupation-measure and moment lifts with nonaffine data entry, per-instance restrictions, and the trivial convex-hull reformulation that every problem admits.

2. *Impossibility* (Section 3). Theorem A: a lossless uniform convex relaxation forces the feasible boundary-data set to be convex. Theorem B and an explicit ball bound: the 5-DOF feasible set is nonconvex, with a gap of metres per second (at boundary data that start below ground for a lander that cannot hover at minimum throttle). Theorem B*: nonconvexity persists with altitude and glide-slope constraints at every horizon, with upright landing, and with free final time on compact windows. A line-search lemma shows that searching over t_f cannot rescue LCvx. Theorem D and Lemma S give per-instance versions for minimum-landing-error problems, with a Wirtinger-type inequality that handles coning manoeuvres. Theorem A-opt and Proposition T give the optimization version. Results for $\rho_1 = 0$ show that the obstruction does not depend on a minimum throttle. Transfers to kinematic and independent-torque 6-DOF models are given. For the real pure-TVC actuation (the gimbal is the pitch/yaw torque), landing upright at rest with altitude and glide-slope constraints, a computer-assisted theorem, certified in exact arithmetic, proves nonconvexity at the vehicle's constants for horizons of 1 and 2 s without a tilt cap, and of 3 and 5 s in the tilt-capped class (constant mass, unlimited gimbal rate).

3. *Possibility* (Section 4). LCvx is exact for any fixed attitude profile. For the gimbal-torque-coupled TVC body with constant mass the problem with a given attitude is exactly convex, with no relaxation. There is an exact subclass for the kinematic model. For degree-2 SDP lifts we give stress certificates and a certified-global coplanar subclass, together with counterexamples to exactness in general. Finally, for the pure-TVC body with depleting mass we give a single SOCP in centre-of-percussion coordinates, with the actuator limits as convex restrictions on Bézier control tuples, and prove that all its feasible points are trajectories of the model within the limits at every instant (Theorem 4.7); by Theorem 3.1 it cannot also be complete (Proposition 4.8).

4. *Implications for guidance* (Section 6). A numerical study on the motivating vehicle confirms the theory's predictions: its 3-DOF guidance reports "optimal" for pad hops no flyable plan can reach, and lands 28% of in-air closed-loop runs, while planners that iterate on the attitude (standard 6-DOF successive convexification, and a planner built directly on the exact convex inner problem) reach the 6-DOF optimum to within 0.03% and land every in-air run. The exact inner problem does not make the latter better than standard successive convexification. The single program of Theorem 4.7 uses 0.1% more fuel at the operating point, needs 0.04–0.22 s more minimum flight time in the air (up to 0.4 s on hops), and solves in about 30 ms. With each planner's computation time counted it lands all 70 test cases, within 1.1 cm in the air against 18 cm for successive convexification; at equal replan rate and zero computation time lateral accuracy is a tie (numerical evidence).

5. *Numerics* (Section 5). Gap laws, results beyond the proven regimes, and the size of the obstruction for a small lander: centimetre-scale for in-air descents, a metre-scale hole in the landing set for pad-launched hops.

6. *Formalization* (Section 7). The Main Theorem above (state-constrained Theorem B* with Theorem A, in 5-DOF and kinematic 6-DOF, with free or upright terminal attitude) and its extensions to mass depletion, free-time windows, the quantitative gap and the vehicle's constants; Theorem B with explicit mass depletion; the centre-of-percussion identity and the logical skeleton of the pure-TVC certificate; the abstract Theorem D, the line-search lemma, the Wirtinger lemma and master inequality, Theorem A-opt, and an explicit instance of Lemma S with the resulting hard-landing Theorem D are machine-checked in Lean 4 with Mathlib [29, 30]. Theorem 4.7 is not formalized beyond its first step, the centre-of-percussion identity. External theorems used on the positive side (the maximum principle and Aumann's theorem) enter only as explicit hypotheses.

Scope, honestly stated. The impossibility concerns the class of convex programs in which the physical state is a decision variable and the boundary data enter affinely. That covers every LCvx formulation in the literature that we know of (Section 2.5). It does not cover convex programs over measures with point-mass data entry, which are exact but infinite-dimensional and per instance, and it does not say that 5-DOF guidance is intractable, and it does not exclude convex programs that are sound without being complete (Section 4.5). For the in-air descents of the motivating vehicle, the nonconvexity we can exhibit is real but thin. For the actual pure-TVC actuation with an upright terminal attitude, the rigorous results cover $t_f = 1,2$ s uncapped and $t_f = 3,5$ s with a tilt cap, all with constant mass and unlimited gimbal rate (Theorem 3.22). Mass depletion, gimbal-rate limits, and longer uncapped horizons remain numerical.

Notation. $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ are Euclidean, S^2 is the unit sphere, e_z points up, $x_z = \langle x, e_z\rangle$ and $x_h = x - x_z e_z$. For unit vectors, $\angle(x,y) = \arccos\langle x,y\rangle \in [0,\pi]$. Gravity is $g = -g_0 e_z$ unless stated otherwise. All functions live on $[0,t_f]$ with Lebesgue measure.

2 Problem formulation

2.1 Translational dynamics and thrust

All models share the translational part

$$\dot r = v, \qquad \dot v = a(t)\,d(t) + g, \qquad \dot m = -\alpha T, \qquad a = T/m, \tag{1}$$

with thrust magnitude $T(t) \in [\rho_1, \rho_2]$ (measurable), $0 < \rho_1 \le \rho_2$ unless $\rho_1 = 0$ is stated, mass-depletion constant $\alpha \ge 0$, initial mass m_0 with $m_0 - \alpha\rho_2 t_f > 0$, and thrust direction $d(t) \in S^2$ (measurable). Integrating (1) gives

$$v(t_f) = v_0 + g t_f + I, \quad r(t_f) = r_0 + v_0 t_f + \tfrac{1}{2} g t_f^2 + J, \qquad I = \int_0^{t_f} a\,d\,ds, \quad J = \int_0^{t_f} (t_f - s)\,a\,d\,ds. \tag{2}$$

Since $T \ge \rho_1$ forces $m(s) \le m_0 - \alpha\rho_1 s$, every admissible acceleration satisfies $a(s) \ge a_1(s) := \rho_1/(m_0 - \alpha\rho_1 s)$, the acceleration at constant minimum throttle.

2.2 Attitude models

The body (thrust) axis is $b(t) \in S^2$. The gimbal constraint is $\angle(d(t), b(t)) \le \delta$ with $\delta \in [0, \pi/2)$. A body-fixed box gimbal $\pm\delta$ per axis is handled by replacing δ by the corner angle δ^* in obstruction arguments ($\delta^* = 21.09°$ for a $\pm 15°$ Cardan gimbal) and by the inscribed angle where a witness needs it .

M0 (3-DOF LCvx baseline).
No attitude; optional pointing cone $\angle(d, \hat{n}) \le \theta_p$ about a fixed axis.

M1 (5-DOF kinematic: rate-limited axis, gimbal cone).
b is continuous, $b(0) = e_z$, and *angle-Lipschitz*:

$$\langle b(s), b(s') \rangle \;\ge\; \cos(\omega |s - s'|) \qquad \text{whenever } \omega |s - s'| \le \pi. \tag{AL$_\omega$}$$

(AL$_\omega$) holds iff b is absolutely continuous with $\dot{b} = \Omega \times b$ and $\|\Omega_\perp\| \le \omega$, where $\Omega_\perp$ is the transverse body rate; roll is unrestricted . The gimbal is fast (d measurable). Two terminal variants: (F) $b(t_f)$ free; (T) $b(t_f) = e_z$.

M3a / M4-dyn (rigid body, independent transverse torque).
Axisymmetric inertia $J = \mathrm{diag}(J_t, J_t, J_a)$, Euler's equations, transverse torque $\|\tau_\perp\| \le \tau_t$ from an actuator independent of the main engine (reaction-control system or wheels), any roll torque, $\Omega(0) = 0$. Optional rate limit.

M4-kin (kinematic 6-DOF).
Full attitude $\mathsf{R}(t) \in SO(3)$ with angular velocity as control, $\|\Omega_\perp\| \le \omega$, arbitrary roll; thrust axis $b = \mathsf{R} e_3$.

PT / M3b (pure TVC rigid body; the realistic small-lander model).
Pitch and yaw torque come only from the gimbal: with the pivot at $-Lb$ from the centre of mass, $\tau_\perp = -LT\, b \times d$. Deflecting the thrust outward rotates the body *inward*. Roll is held by a separate RCS.

Roll does not move the thrust axis of an axisymmetric vehicle. 5-DOF (translation plus pitch and yaw of b) is therefore the natural primary model, and 6-DOF results follow by projection and lifting (Section 3.8).

Lemma 2.1 (Centre-of-percussion identity)**.** *Consider model PT with constant mass m, axisymmetric J, roll rate $\varpi = \langle \Omega, b \rangle$ and $\varepsilon := J_t/(mL)$. The point $y = r + \varepsilon b$ satisfies*

$$\ddot{y} = \Big(\frac{T \cos\mu}{m} - \varepsilon \|\Omega_\perp\|^2\Big) b + g + \frac{J_a}{mL}\, \varpi\, \Omega_\perp, \qquad \mu = \angle(d, b).$$

With zero roll rate the lateral gimbal force cancels exactly: y is driven by a thrust along b (effective gimbal 0) whose magnitude is reduced by the centripetal term. [proved; not machine-checked] [Lean-checked: zero roll rate and with roll rate; constant ε]

For the motivating vehicle, $\varepsilon \approx 1.35$ m. The identity is the basis of the pure-TVC obstruction (Section 3.8.1), of the pure-TVC positive result (Theorem 4.4), and of the sound single convex program of Section 4.5, where y is used as a flat output.

2.3 Boundary data and feasible sets

The boundary data are $\theta = (r_0, v_0) \in \mathbb{R}^6$. The mass m_0, the initial attitude (and initial rate in torque models), the terminal attitude in variant (T), and t_f are fixed. Landing means $r(t_f) = 0$, $v(t_f) = 0$.

Definition 2.2 (Feasible boundary-data sets)**.** $F_{t_f} = \{\theta : \text{some admissible control lands from } \theta \text{ at } t_f\}$. For a closed state-constraint set $\mathcal{X} \subseteq \mathbb{R}^6$, $F_{t_f}^{\mathcal{X}}$ is the subset whose landing trajectory satisfies $(r(t), v(t)) \in \mathcal{X}$ for all t. Our main choice is altitude plus glide slope,

$$\mathcal{X} = \{(r, v) : \ r_z \geq 0, \ r_z \geq \tan\gamma_{gs}\|r_h\|\}, \qquad \gamma_{gs} \in [0, \pi/2). \tag{3}$$

The *velocity-increment set* is $R = \{I\}$ over admissible controls, and $K = \{(I, J)\}$.

By (2), $F_{t_f} = A^{-1}(K)$ for an affine bijection A, and $\pi_v(F_{t_f}) = -gt_f - R$. Hence

$$F_{t_f} \text{ convex} \iff K \text{ convex} \implies R \text{ convex}.$$

For one fixed instance θ_0 we also use $\mathrm{Reach}(\theta_0)$, the set of terminal states of admissible trajectories that stay in $\mathcal{X}$, and the *landing-offset set* $L(\theta_0) = \{p : \text{some admissible trajectory from } \theta_0 \text{ ends at } r(t_f) = (p, 0), \ v(t_f) = 0\}$.

2.4 Relaxations and losslessness

Definition 2.3 (Uniform convex relaxation)**.** Let $\Theta = \mathbb{R}^6$ and let Y be any real vector space (lifted variables, discretized controls, slacks; possibly infinite-dimensional). A *uniform convex relaxation* is a convex set $C \subseteq \Theta \times Y$. Its slice $C_\theta = \{y : (\theta, y) \in C\}$ is the feasible set of the relaxed problem at θ, which may minimize any cost. The data enter *affinely* in the sense that C is convex jointly in (θ, y). More generally, they enter through an affine bijection ψ of Θ, or through a *separable monotone* map $\psi(\theta) = (\psi_r(r_0), \psi_1(v_{0x}), \psi_2(v_{0y}), \psi_3(v_{0z}))$ with each ψ_i continuous and strictly monotone in a frame with vertical third axis.

Definition 2.4 (Losslessness)**.** (a) *Feasibility (detection-sound) form:* C is lossless for $F \subseteq \Theta$ if $C_\theta \neq \emptyset \iff \theta \in F$ for every θ. It is *ε-lossless* if $F \subseteq \pi_\Theta(C) \subseteq \{\theta : \mathrm{dist}(\theta, F) \leq \varepsilon\}$. (b) *Optimization (LCvx) form* for a cost J: every original-feasible θ has $C_\theta \neq \emptyset$, and whenever $C_\theta \neq \emptyset$ the relaxed minimum over C_θ is attained at a point that certifies an original-feasible (and optimal) solution at θ. (c) *Value-lossless on F:* $\min_{C_\theta} J = V(\theta)$ for every $\theta \in F$, where V is the original value.

Proposition 2.5 (LCvx form implies feasibility form)**.** *If C is lossless in the optimization form (b) for some cost, it is lossless in the feasibility form (a).* [Lean-checked]

Proof. The first half of (b) gives "$\Leftarrow$". If $C_\theta \neq \emptyset$, the attained relaxed optimum certifies an original-feasible solution, so $\theta \in F$. □

So the feasibility form is not stronger than the definition used in the LCvx literature [4, 14], provided relaxed optima are attained (which thrust bounds guarantee). A relaxation that is exact on F but reports false feasibility off F fails (b). That is the failure mode Theorem A forces, and the optimization form (c) is treated separately in Section 3.6.

Definition 2.6 (State-linear relaxation of one instance)**.** Fix an instance θ_0. A *state-linear relaxation* is (Y, C, ρ, P) with $C \subseteq Y$ convex, $\rho : Y \to \mathbb{R}^6$ affine (the physical readout of the terminal state, or of (r, v) at finitely many times), and $P : Y \to [0, \infty]$ convex (a penalty), such that every genuine trajectory with terminal state x_f is represented by some $y \in C$ with $\rho(y) = x_f$ and $P(y) = 0$. For a convex terminal-error function φ, the relaxed problem is $\inf_{y \in C} \varphi(\rho(y)) + P(y)$; the original is $\inf_{x \in \mathrm{Reach}(\theta_0)} \varphi(x)$.

2.5 Why this class captures LCvx, and what it excludes

Every LCvx formulation we know enters (r_0, v_0) through linear equality constraints $x(0) = x_0$ on top of a convex constraint set in the remaining variables (states, relaxed controls, slack Γ, discretization). It is therefore a uniform convex relaxation at fixed t_f. The log-mass change of variables $z = \ln m$ [2] acts on Y, and the only data reparametrization in the literature is coordinatewise monotone. The cost of LCvx (fuel, landing error) is a convex function of the physical variables plus slack penalties that vanish on genuine points, which is Definition 2.6. Statements are made in continuous time. Discrete-time LCvx is not exact in general either [12], and continuous-time statements avoid conflating the two effects.

The class excludes the following on purpose. We consider these the honest boundary of the impossibility.

1. *Nonaffine data entry.* Any compact $S \subset \Theta$ is the preimage of a convex set of measures under $\theta \mapsto \delta_\theta$ (point mass). Without a restriction on how data enter, convexity of a feasible set has no content.
2. *Occupation-measure and moment-SOS lifts* [31, 32]. Here the initial state enters as a point mass, terminal conditions act on the support of a lifted measure, and costs are integrals against it. These programs are convex and, under standard conditions, value-exact [33, 34], but they are infinite-dimensional and set up per instance. Their order-one truncation is exactly the affine case and inherits the obstruction. Degree-2 truncations are studied in Section 4.6.
3. *Costs on lifted second moments* (e.g. $R \succeq r(t_f) r(t_f)^\top$ with cost $\mathrm{tr}\, R - 2q^\top r$). On mixtures these average the error instead of evaluating it at the mean readout.
4. *Per-instance restrictions* such as a fixed attitude profile. These violate the embedding property of Definition 2.6, and they are the positive side of the story (Section 4).

One kind of program is *inside* the class and still useful: a uniform convex *restriction*, which is feasible only at data in F but not at all of it. Definition 2.4(a) asks for both directions, soundness and completeness, and Theorem 3.1 shows the two cannot hold together when F is nonconvex. Soundness alone is compatible with it. Section 4.5 constructs such a program for model PT.

Proposition 2.7 (Trivial exact convex reformulation)**.** *Every optimal control problem* $\min_{x \in \mathcal{F}} J(x)$ *with* J *bounded below has an exact convex reformulation: minimize the linear functional* $(x, t) \mapsto t$ *over* $\mathrm{conv}\{(x, J(x)) : x \in \mathcal{F}\}$*. If the graph is compact, the minimum is attained at a genuine solution.* *[proof sketch only]*

This reformulation is per instance, not state-linear (the cost is a lifted coordinate that averages over mixtures), and describing the hull is as hard as the original problem. "Is LCvx possible?" is therefore a meaningful question only inside a tractable class. Our answer is the statement that the state-linear class cannot be exact.

3 Impossibility results

3.1 The abstract obstruction

Theorem 3.1 (Theorem A)**.** *If a uniform convex relaxation C with affine data entry is lossless for F (feasibility form), then $F = \pi_\Theta(C)$; in particular F is convex. Consequently, if F is not convex, no uniform convex relaxation is lossless for F, in either the feasibility or the optimization form, whatever the cost.* [Lean-checked]

Proof. $\theta \in \pi_\Theta(C) \iff C_\theta \neq \emptyset \iff \theta \in F$, and a linear image of a convex set is convex. The optimization form reduces to the feasibility form by Proposition 2.5. □

Corollary 3.2. *(i) If the data enter through an affine bijection ψ, then $\psi(F)$, hence F, is convex. (ii) If C contains the original problem ($F \subseteq \pi_\Theta(C)$), then $\pi_\Theta(C) \supseteq \operatorname{conv} F$, so C reports feasibility on all of $\operatorname{conv} F \setminus F$. If $\operatorname{dist}(\theta_M, F) = \Gamma > 0$ for some $\theta_M \in \operatorname{conv} F$, no uniform convex relaxation is ε-lossless for $\varepsilon < \Gamma$.* [Lean-checked] *(part (i))*

Theorem A is elementary. The substance lies in showing that the 5-DOF and 6-DOF feasible sets are not convex, in the physically relevant settings, and in establishing that the definitions match LCvx (Section 2.5). Theorem A also explains why 3-DOF LCvx is consistent: for the exact 3-DOF problem F_{t_f} is convex by Aumann's theorem [28] (Remark 3.23).

Lemma 3.3 (Reduction to velocity increments)**.** *Let $P_\pm \in R$ be realized by controls landing from $\theta_\pm$, $M = \frac{1}{2}(P_+ + P_-)$ and $\theta_M = \frac{1}{2}(\theta_+ + \theta_-) \in \operatorname{conv} F_{t_f}$. Then $\operatorname{dist}(\theta_M, F_{t_f}) \geq \operatorname{dist}(M, R)$. In particular $M \notin R \Rightarrow \theta_M \notin F_{t_f}$.* [Lean-checked]

3.2 Theorem B: the unconstrained 5-DOF feasible set is nonconvex

Let $\varphi_F(s) = \omega s$ (variant F) or $\varphi_T(s) = \omega \min(s, t_f - s)$ (variant T). For a horizontal unit vector u, the *extreme arc* tilts the body at full rate toward u, deflects the thrust a further δ outward, and holds minimum throttle:

$$b_u = \cos\varphi\, e_z + \sin\varphi\, u, \quad d_u = \cos(\delta+\varphi)\, e_z + \sin(\delta+\varphi)\, u, \quad T \equiv \rho_1. \tag{4}$$

Its velocity increment is $P_u = (\int a_1 \sin(\delta+\varphi))u + (\int a_1 \cos(\delta+\varphi))e_z$.

Theorem 3.4 (Theorem B)**.** *Model M1 with $\rho_1 > 0$, $0 < \omega < \infty$, $\delta \in [0, \pi/2)$, $\alpha \geq 0$, any g, and either (F) $b(t_f)$ free with $\delta + \omega t_f \leq \pi/2$, or (T) $b(t_f) = e_z$ with $\delta + \omega t_f/2 \leq \pi/2$. Then $M = \frac{1}{2}(P_u + P_{-u}) = (\int_0^{t_f} a_1 \cos(\delta+\varphi))\, e_z$ lies in $\operatorname{conv} R \setminus R$. Hence R and F_{t_f} are nonconvex, and no uniform convex relaxation of the 5-DOF landing problem is lossless.* [Lean-checked: variant F, any continuous $a_1 > 0$; explicit mass-depletion model] [proved; not machine-checked] *(variant T)*

Proof sketch (full proof in Appendix A). Suppose $M = \int a\, d$ for an admissible control. (1) By (AL_ω) from $s' = 0$, $b_z(s) \geq \cos\varphi(s)$. The spherical triangle inequality and the gimbal constraint give $d_z \geq \cos(\delta+\varphi) \geq 0$, so $a\, d_z \geq a_1 \cos(\delta+\varphi)$ pointwise. (2) Integrating, $M_z \geq \int a_1 \cos(\delta+\varphi) = M_z$, so equality holds a.e. (3) Equality in the triangle inequality forces $b_z = \cos\varphi$ and puts d in the vertical plane of b, tilted outward: $d = \cos(\delta+\varphi)e_z + \sin(\delta+\varphi)u(s)$ with $u(s)$ the azimuth of $b(s)$. (4) *Rigidity:* for $s < s'$ on a monotone piece of φ, (AL_ω) gives $\langle b(s), b(s')\rangle \geq \cos(\varphi(s') - \varphi(s))$, which with both tilts exact forces $u(s) = u(s')$. So the azimuth is constant. (5) Then $M_h = (\int a \sin(\delta+\varphi))u_0 \neq 0$, a contradiction. □

Step (4) is where the finite rate enters. With $\omega = \infty$ the azimuth can chatter between $\pm u$ and $M \in R$, which is consistent with 3-DOF LCvx. With $\omega = 0$ the attitude is fixed and $M \in R$ again, by chattering of the thrust azimuth on the gimbal cone. So $\omega \in (0, \infty)$ is essential. The gimbal is not: $\delta = 0$ works.

Quantitative gap: the ball bound

Theorem 3.5 (Theorem Q, ball bound). *In M1, variant F, assume* $\delta + \omega t_f/2 \le \pi/2$ *and* $M_z = \int a_1 \cos(\delta + \omega s)\, ds \ge 0$ *(automatic if* $\alpha = 0$ *or* $\delta + \omega t_f \le \pi/2$*). Every* $I \in R$ *satisfies* $\|I\| \ge \langle I, b(t_f/2)\rangle \ge G(t_f/2)$*, where* $G(\tau) = \int_0^{t_f} a_1 \cos(\delta + \omega|s - \tau|)\, ds$*. Consequently*

$$\operatorname{dist}(\theta_M, F_{t_f}) \;\ge\; \operatorname{dist}(M, R) \;\ge\; G(t_f/2) - M_z \;\ge\; \Gamma_F := \frac{4a_1(t_f/4)}{\omega} \sin\Big(\delta + \frac{\omega t_f}{2}\Big) \sin^2\Big(\frac{\omega t_f}{4}\Big) > 0,$$

with equality in the last step if $\alpha = 0$*, and no uniform convex relaxation is* ε*-lossless for* $\varepsilon < \Gamma_F$*.* [proved; not machine-checked]

The proof needs only the triangle inequality, without its equality case, and one explicit integral (Appendix A). For $\rho_1 = 400$ N, $m_0 = 80$ kg, $\delta = 15°$, $\omega = 0.3$ rad/s, $t_f = 5$ s the bound is 7.6 m/s. Numerically it is within 17–19% of the true distance over 39 configurations (Section 5). For small ωt_f, $\Gamma_F \approx \frac{1}{4} a_1 \omega t_f^2 \sin\delta$ ($\delta > 0$), so the gap vanishes as $\omega \to 0$ and is linear in ω. A two-reference-direction refinement gives an explicit gap $\gamma_T > 0$ for the tent variant (T) when $\alpha = 0$; it is a min–max formula valid only under the hypothesis $\delta + \omega t_f/2 \le \pi/2$ (e.g. $\gamma_T \approx 3.24$ m/s at $t_f = 6$ s, $\omega = 0.3$ rad/s, $\delta = 15°$, $\rho_1 = 400$ N, $m_0 = 80$ kg). Outside that hypothesis the formula is not a valid bound, and nonconvexity is covered only qualitatively by the all-horizon result below.

Proposition 3.6 (All horizons, free terminal attitude). *In M1 with* $\rho_1 > 0$*,* $0 < \omega < \infty$*,* $0 \le \delta < \pi$*,* $\alpha \ge 0$ *and* $b(t_f)$ *free,* F_{t_f} *is nonconvex for* every $t_f > 0$*. With* $\rho_1 = 0$ *and* $\alpha = 0$ *the same holds whenever* $\delta + \omega t_f > \pi/2$*.* [proved; not machine-checked]

The witnesses of Proposition 3.6 use capped arcs (and inverted thrust when $\delta + \omega s > \pi/2$), so their physical content lies mostly in Theorems 3.4 and 3.5. They do show that the obstruction is not an artefact of short horizons.

3.3 State constraints: the dive-then-brake face (Theorem B*)

The witnesses of Theorems B and Q fly at minimum throttle throughout. A small lander cannot hover at minimum throttle ($\rho_1/(m_0 g_0) \approx 0.51$ for our vehicle), so these witnesses start *below ground*. Theorems B and Q say nothing about the altitude-constrained problem. The repair uses a different exposed face. Fix a switch time $s^* \in (0, t_f)$ and the rotation-invariant linear functional $\ell(\theta) = r_{0z} + s^* v_{0z}$. A landing control satisfies the exact identity

$$\Lambda(\theta) := r_0 + s^* v_0 = \int_0^{t_f} (s - s^*)\, a\, d\, ds + g_0 t_f (s^* - \tfrac{t_f}{2}) e_z. \tag{5}$$

The weight $s - s^*$ is negative before s^* and positive after. Maximizing ℓ therefore asks for a *dive* (minimum throttle, tilted as far as possible) followed by a *brake* (full throttle, vertical thrust). The brake keeps the trajectory above ground.

Theorem 3.7 (Theorem B*). *Model M1 with* $\rho_1 > 0$*,* $0 < \omega < \infty$*,* $0 < \delta < \pi/2$*,* $\alpha \ge 0$*,* $A_b := \rho_2/m_0 - g_0 > 0$*, and the state constraint* (3)*. Let* $s^* \in (0, t_f)$ *satisfy*

(S1) $\omega s^* \le \min(\delta, \frac{\pi}{2} - \delta)$, *(S2)* $g_0 s^* < A_b(t_f - s^*)$, *(S3)* $\tan\gamma_{gs}\, a_{\max} s^{*2} \le A_b (t_f - s^*)^2$.

Then there are $\theta^* \in F_{t_f}^{\mathcal{X}}$ and its rotation $R_\pi\theta^* \in F_{t_f}^{\mathcal{X}}$ by π about e_z whose midpoint θ_m (with zero lateral components) is not in the unconstrained *set F_{t_f}. Hence $F_{t_f}^{\mathcal{X}}$ is nonconvex for every $t_f > 0$ (for every t_f a small enough s^* exists), and no uniform convex relaxation of the altitude- and glide-slope-constrained 5-DOF problem is lossless. The same proof covers: (T′) upright landing $b(t_f) = e_z$ when $t_f \geq 2s^*$; free final time on any compact window $[t_a, t_b]$, i.e. $\bigcup_{t\in[t_a,t_b]} F_t^{\mathcal{X}}$ is nonconvex; any additional closed, R_π-invariant constraint satisfied by the witness (e.g. a tilt cap $\geq \omega s^*$). If moreover $\delta \leq \pi/4$, then*

$$\operatorname{dist}(\theta_m, F_{t_f}) \;\geq\; \frac{\eta_0}{\sqrt{1+s^{*2}}}, \qquad (1+c_1)\eta_0 + \sqrt{2W_b\eta_0} = \gamma_X,$$

with $\gamma_X = \frac{5}{32}a_{10}\sin\delta\, s^{*2}$, $c_1 = 4a_{\max}/(\omega a_{10}\sin\delta\, s^*)$, $W_b = \frac{1}{2}a_{\max}(t_f - s^*)^2$, $a_{10} = \rho_1/m_0$, $a_{\max} = \rho_2/(m_0 - \alpha\rho_2 t_f)$. [Lean-checked: every t_f, any $\mathcal{X} \supseteq$ glide-slope set, $b(t_f)$ free and upright (T′), 5-DOF and kinematic 6-DOF, constant bounds and explicit mass depletion; free-time windows; quantitative bound in max-norm form] [proved; not machine-checked] *(all variants and the quantitative bound)*

Proof sketch (full proof in Appendix B). For a fixed throttle, $W_z = \int (s - s^*)a\, d_z$ is bounded above pointwise: on the dive $d_z \geq \cos(\delta + \omega s)$ and the weight is negative; on the brake $d_z \leq 1$. One attitude history attains both bounds. It dives at full rate for s^* and then holds its tilt $\omega s^* \leq \delta$, so vertical thrust remains inside the gimbal cone. Full throttle on the brake is optimal by a pointwise mass comparison. The witness pair lands from θ^* and $R_\pi\theta^*$. It stays above ground, because the brake decelerates at $\geq A_b$, and inside the glide slope by (S3). If θ_m were feasible, its control would attain the bound, so the equality case and the rigidity argument of Theorem B apply on the dive and force $d = e_z$ on the brake. Its weighted lateral moment would be nonzero, whereas $\Lambda_h(\theta_m) = 0$. The quantitative part avoids the equality case. It uses a support bound at $\tau = \vartheta s^*$ for the dive and Cauchy–Schwarz for brake-phase tilts. □

Remark 3.8 (Formal version and its hypotheses). This is the paper's headline machine-checked result (Section 7). The Lean statement `not_convex_constrainedLandingSet` assumes constant acceleration bounds $0 < a_{\min} \leq a \leq a_{\max}$ (constant mass), $g_0 \leq a_{\max}$, $\omega s^* \leq \delta$, $\delta + \omega s^* \leq \pi/2$, and the single smallness condition $(a_{\max} + 2\tan\gamma\, a_{\min})s^{*2} \leq (a_{\max} - g_0)(t_f - s^*)^2$ in place of (S2)–(S3). `exists_switchTime` shows that such an s^* exists for every $t_f > 0$, $0 < \delta < \pi/2$, $g_0 < a_{\max}$. The statement holds for every $\mathcal{X}$ containing the glide-slope set. The upright-landing variant (T′, with $2s^* \leq t_f$, also supplied by `exists_switchTime`) is `not_convex_constrainedLandingSetUp`. The kinematic 6-DOF versions (attitude a path of proper rotations, total body rate $\leq \omega$, roll angle free) are `no_lossless_relaxation_constrained_6dof` and `no_lossless_relaxation_constrained_upright_6dof`. With explicit mass depletion (model `AccMass`) the headline is `no_lossless_relaxation_constrained_5dof_mass`, with upright and kinematic 6-DOF variants. Free final time on compact windows is `no_lossless_relaxation_freeTime_5dof` (and `_mass`). The quantitative bound is `landingSet_dist_ge` and `midpoint_not_mem_closure_landingSet`, with `landingSet_dist_ge_mass` for mass depletion; it uses the max-norm on $\mathbb{R}^3 \times \mathbb{R}^3$, so its constant reads $\eta_0/(1+s^*)$. The instances at the vehicle's constants are `gtpl_no_lossless_upright_6dof_mass` and `gtpl_no_lossless_freeTime_mass`. The envelope version B*-env is not formalized.

Remark 3.9 (The B* gap is thin, and why). For the motivating vehicle ($\rho_1 = 400$ N, $\rho_2 = 1000$ N, $m_0 = 80$ kg, $\delta = 15°$, $\omega = 0.4$ rad/s, $\gamma_{gs} = 30°$, $s^* = \delta/\omega$, $t_f = 7$ s), $\gamma_X = 0.087$ m but $\eta_0 \approx 1.3 \times 10^{-5}$ m. This is not slack in the proof. A competitor can cancel the dive's lateral moment by tilting slightly during the long brake, at a cost quadratic in the tilt, of order $\|X_h\|^2/(2W_b)$. Dropping $\omega s^* \leq \delta$ gives an envelope version (Theorem B*-env) whose heuristic gap is at most about 1 cm. Numerically the B* face gap is about 1 cm/s (Section 5). *So the nonconvexity of the real*

constrained problem is proved, but a practically significant ε-impossibility for in-air descents of this vehicle is not.

3.4 Free final time and line search

LCvx usually treats free final time by a line search over t_f, solving one convex program per horizon [4, 5]. Two routes are possible. A single program in which t_f is a jointly convex decision variable fails immediately: every slice F_t would then be convex . Line search needs an argument, because the accepted set is a union of convex sets.

Lemma 3.10 (Line search cannot be lossless; RR-LS)**.** *Let W be a set of horizons, F_t the true feasible sets, and $\{C_t\}_{t\in W}$ per-horizon uniform convex relaxations with $F_t \subseteq \pi_\Theta(C_t)$. (i) If some $\bar{x} \in \operatorname{conv} F_{t^*}$, $t^* \in W$, lies in no F_t, $t \in W$, then line search accepts $\bar{x}$ although it is infeasible at every horizon. (ii) Such an $\bar{x}$ exists if there are a linear involution R preserving every F_t and an R-invariant linear functional ℓ with a maximizer over the graph $\{(t, x) : t \in W, x \in F_t\}$, and every maximizer has $x \neq Rx$.* [Lean-checked: abstract (i)–(ii)]

Corollary 3.11. *Let $W = [t_a, t_b]$ with $s^* < t_a$, such that (S1)–(S3) of Theorem 3.7 hold at $t_f = t_a$ with $a_{\max}$ evaluated at t_b. If a dry-mass constraint is imposed, assume also $m_0 - \alpha\rho_2 t_b \geq m_{dry}$ (the B^* witness brakes at full throttle). Then the B^* functional ℓ and the rotation R_π supply (ii): $\ell^*(t) = \sup_{F_t} \ell$ is continuous and attained, and every maximizer has $\Lambda_h \neq 0$. Hence no per-horizon family of uniform convex relaxations makes free-final-time 5-DOF landing lossless on W, in feasibility or optimization form.* [proved; not machine-checked]

A window defined through the minimum-throttle fuel time $(m_0 - m_{dry})/(\alpha\rho_1)$ is *not* valid here, because on horizons beyond $(m_0 - m_{dry})/(\alpha\rho_2)$ the full-throttle brake of the witness runs out of fuel. Some window is necessary for this witness: the unconstrained midpoint of Theorem B becomes reachable at a longer horizon when $g_0 > a_1 \cos\delta$, and over unbounded horizons $\sup \ell = \infty$.

3.5 Per-instance obstructions: Theorem D and Lemma S

Theorem A concerns relaxations that are uniform in the data. A guidance computer solves one instance at a time, so one may ask whether some convex program is exact for each given instance, possibly exploiting a rotated frame or symmetry. Definition 2.6 answers this for minimum-landing-error problems.

Theorem 3.12 (Theorem D)**.** *For any state-linear relaxation of an instance, $\rho(\{y \in C : P(y) = 0\}) \supseteq \operatorname{conv}\operatorname{Reach}(\theta_0)$. Hence the relaxed value is at most $\inf_{\operatorname{conv}\operatorname{Reach}} \varphi$. If $q \in \operatorname{conv}\operatorname{Reach} \setminus \operatorname{cl}\operatorname{Reach}$ and $\varphi = \|\cdot - q\|$, the relaxed value is 0, the true value is $\operatorname{dist}(q, \operatorname{Reach}) > 0$, and every relaxed optimum reads out a non-genuine state.* [Lean-checked]

Proof. Mix the embedded points y_i of genuine states x_i. The readout is affine and the penalty convex and vanishing at the y_i. □

Theorem D needs no joint convexity in the data and is immune to frame rotation. It is useful only once a nonconvex reachable set is exhibited at a *single* feasible instance. With a soft terminal velocity, the vertical instance θ_m of Theorem B* gives one, with a certified (tiny) gap when $\delta \leq \pi/4$. Additional constraints on the vertical state (e.g. descent-rate limits) are allowed only if the B* witness satisfies them. The LCvx soft-landing problem imposes $v(t_f) = 0$ hard and penalizes the landing offset. Theorem D then needs a nonconvex landing-offset set:

(S) there is vertical data $\zeta = (r_0 = he_z,\ v_0 = v_{0z}e_z)$ with $0 \notin \operatorname{cl} L(\zeta) \neq \emptyset$.

$L(\zeta)$ is rotation invariant, so (S) says it is an annulus with a hole around the point directly below. The danger is *coning*: precessing the azimuth at fixed tilt cancels lateral motion and might fill the hole. A Wirtinger inequality modulo affine functions handles all such manoeuvres at once.

Lemma 3.13 (Wirtinger modulo affine functions)**.** *Let P be the $L^2(0,T)$-orthogonal projection onto affine functions. For $f \in H^1((0,T);\mathbb{R}^N)$,*

$$\|f - Pf\|_{L^2}^2 \le \Big(\frac{T}{2\pi}\Big)^2 \Big(\|\dot f\|_{L^2}^2 - \frac{|f(T) - f(0)|^2}{T}\Big),$$

with equality for $f = c(1-\cos(2\pi s/T))$. In particular an ω-Lipschitz path β (of any range dimension) satisfies $\|\beta - P\beta\| \le \omega T^{3/2}/(2\pi)$. [proved; not machine-checked] [Lean-checked]

The inequality is componentwise, so planar S-turns, coning paths and even $\mathbb{R}^N$-valued (lifted) lateral paths obey the same bound.

Theorem 3.14 (Theorem S: Lemma S in an explicit regime)**.** *Model M1, $b(t_f)$ free, $\alpha \ge 0$, $g = -g_0 e_z$. Let $\bar\tau = \delta + \omega t_f$ and assume (S0) $\bar\tau < \pi/2$, (S1) $(\rho_1/m_0)\cos(\delta + \omega t_f) > g_0$ (minimum-throttle thrust-to-weight above $1/\cos\bar\tau$; altitude is then automatic), (S2) a throttle condition that is automatic if $\rho_1 = \rho_2$, and (S3) $M := \mathrm{gain}_w - B > 0$, where gain_w is the vertical-impulse waste of an explicit planar single-crossing S-turn witness and B is an explicit constant (Appendix C). Then at the witness's vertical data ζ_w, $L(\zeta_w)$ contains a full circle, and every landing offset satisfies*

$$|p| \ \ge\ r_S = \Big(\frac{M t_f^3}{12\, C_m}\Big)^{1/2} > 0.$$

Hence (S) holds, and every state-linear relaxation of the minimum-landing-error problem from ζ_w with hard $v(t_f) = 0$ and target $q = 0$ has relaxed value 0 while the true value is at least r_S. [proved; not machine-checked] [Lean-checked: master inequality (constant mass); Lemma S and hard-landing Theorem D for one explicit instance]

Machine-checked instance. For fixed thrust $a \equiv 12$ m/s^2, $g_0 = 9.81$, $\delta = 0$, $\omega = 0.1$ rad/s, $t_f \approx 3.50$ s ($\omega t_f \approx 0.35$), with $r_z \ge 0$ and every admissible 3-D attitude path (coning included), Lean proves that every landing offset has $|p| \ge 1$ m, that $0 \in \operatorname{conv} L \setminus \operatorname{cl} L$ (so L is not convex), and that every state-linear relaxation containing L attains error 0 at $q = 0$ while every genuine landing has error ≥ 1 m (`Slice.lemmaS`, `Slice.theoremD_hard_landing`). The Lean witness is a "hold" variant of the S-turn (tilt up to 0.1 rad, back through zero to -0.1 rad, hold), with the same mechanism. The numerical dent of that instance is about 1.38 m. For a demonstration vehicle with $a_1 = 12$, $a_2 = 25$ m/s^2 and $t_f = 2$ s, the theorem is certified for $0 < \omega t_f \le 0.55$ at $\delta = 0$, for $0.004 \le \omega t_f \le 0.46$ at $\delta = 5°$, and for $0.038 \le \omega t_f \le 0.34$ at $\delta = 10°$, with certified dents $r_S = 0.4$–1.8 m (Table 3). For $\delta > 0$ the range has a positive lower end: as $\omega t_f \to 0$ at fixed δ the witness gain tends to $a_{10} t_f(1 - \cos\delta)$, which is below the limit of B, so $M < 0$. *Realism caveat:* (S1) requires a vehicle that cannot hover at minimum throttle. That excludes essentially every deeply throttleable lander, including ours ($\rho_1/m_0 = 5.0$–7.2 m/s$^2 < g_0$). For our vehicle, (S) holds numerically at dive-then-brake data, with landing-set gaps of about 1 cm in ℓ and optimizers landing 1.5–2.4 m off axis (Section 5). The rigorous GTPL case is open, because the L^2 slack (cm–dm) exceeds the ℓ-gap (mm–cm).

3.6 Optimization form: Theorem A-opt and Proposition T

A weaker notion of losslessness escapes Theorem A: exactness of the optimal value on F only, with no infeasibility detection off F. Mixtures of genuine trajectories constrain it too.

Theorem 3.15 (Theorem A-opt). *Let $C \subseteq \Theta \times Y$ be a uniform convex relaxation with a cost J convex on C, in which every genuine trajectory τ is represented with $J = \mathrm{cost}(\tau)$. If C is value-lossless on F, then $V(\sum_i \lambda_i \theta_i) \le \sum_i \lambda_i V(\theta_i)$ for finite mixtures whenever all θ_i and $\sum \lambda_i \theta_i$ lie in F: on F the value function agrees with its convex envelope over finite mixtures,* $\operatorname{conv} V$. *(Agreement with the closed envelope V^{**} would additionally need lower semicontinuity of the relaxed value.)* [proved; not machine-checked] [Lean-checked]

Proposition 3.16 (Proposition T: an attitude-sensitive cost). *In M1 variant (T) ($b(0) = b(t_f) = e_z$) with $\alpha = 0$, cost $\int_0^{t_f} \langle b, e_z \rangle \, ds$ (maximize mean tilt), $\omega t_f/2 > \delta$ and $a_2 \cos(\omega t_f/2 - \delta) \ge a_1$, there is $\theta' \in F$ at which every such relaxation has value strictly below $V(\theta')$. The same holds in variant (F) with ωt_f in place of $\omega t_f/2$ in both conditions. So no uniform convex relaxation is value-lossless on F for all convex costs.* [proved; not machine-checked] [Lean-checked: abstract implication only; the model facts that supply its hypotheses are not formalized]

For minimum fuel the question is open. Numerically the rotational axis test shows no violation of $V = \operatorname{conv} V$. Even if minimum fuel were value-lossless on F, the relaxation would report spurious feasibility on $\operatorname{conv} F \setminus F$, which is not LCvx in the sense of Definition 2.4(b).

3.7 The minimum throttle is not the cause: $\rho_1 = 0$

Theorems B and B* use the minimum throttle. If the engine may shut down ($\rho_1 = 0$), the velocity set does become convex in the short-horizon regime, but the landing set does not.

Proposition 3.17 (P0-R). *In M1 with $\rho_1 = 0$, $\alpha = 0$, $b(t_f)$ free: R is convex iff $\delta + \omega t_f \le \pi/2$, and then $R = \{\int u : \|u(s)\| \le a_2,\ \angle(u(s), e_z) \le \delta + \omega s\}$, the 3-DOF time-varying-envelope set. In the tent variant (T), R is convex if $\delta + \omega t_f/2 \le \pi/2$ and nonconvex if $\pi/2 < \delta + \omega t_f/2 \le \pi$; beyond that it is open.* [proved; not machine-checked]

Theorem 3.18 (P0-F: landing set nonconvex with $\rho_1 = 0$). *In M1 with $\rho_1 = 0$, $\alpha = 0$, $b(t_f)$ free, $0 \le \delta < \pi/4$ and $0 < \omega t_f < \pi/2 - 2\delta$, the set K, hence F_{t_f}, is nonconvex. No uniform convex relaxation is lossless even with engine shutdown allowed.* [proved; not machine-checked]

Proof idea (Appendix D). On the face $\langle \hat{p}, r_0 + \sigma v_0 \rangle$ with $\hat{p}$ pointing δ below the horizontal, the face value reduces exactly to a one-dimensional maximization over the tilt reached at σ. An intermediate-value argument in σ produces a tie between two separated groups of maximizers. Both "coast while rotating", and the thrust on/off boundary acts as a convex kink. The face is a disjoint union of two nonempty compact sets, so it is not convex. □

With an altitude constraint the same mechanism gives nonconvexity of $F_{t_f}^{\mathcal{X}}$ (early-kink face) when full-throttle thrust-to-weight is at least $1/(\cos(\delta + \omega t_f)(1 - x_*^2))$. This threshold is about 2.1 only as $\omega t_f \to 0$ and increases with ωt_f: at $\delta = 15°$ it is 2.16, 2.38 and 2.78 for $\omega t_f = 0.1$, 0.3 and 0.5. That covers neither our vehicle ($T/W \approx 1.27$) nor glide slopes. The "LCvx is possible iff $\rho_1 = 0$" dichotomy is therefore false: *the obstruction is driven by the finite attitude rate itself.* Proposition 3.17 does give a small exact result: for problems that constrain only the terminal velocity, the 3-DOF envelope SOCP is feasibility-exact when $\rho_1 = 0$.

Two further loopholes are closed. A separable monotone reparametrization of the data cannot convexify F, because the lowest slice of $\pi_v F$ is a circle . The only rate notion that is SOC-representable (a thrust-vector "chord rate" $\|\dot{u}\| \le \omega\sigma$) is not uniformly lossless, because its original problem is again of type M1 with $\delta = 0$. A tilt-angle-only rate limit, with the azimuth free to jump, leaves rigidity broken and appears to give convex landing sets numerically (Section 5); it is, however, not SOC-representable.

3.8 6-DOF

Theorem 3.19 (Transfer to kinematic 6-DOF)**.** *In M4-kin the projection* $(\mathsf{R}, \Omega, T, d) \mapsto (b = \mathsf{R}e_3, T, d)$ *maps admissible 6-DOF controls into admissible M1 controls, and the planar witnesses lift with zero roll, as rotations about a fixed horizontal axis. Hence* $R_6 \subseteq R_5$ *contains the same witnesses, and Theorems B, Q,* B^* *(and their variants) hold verbatim for 6-DOF with fixed attitude data. With a box gimbal, replace* δ *by the corner angle and take* u *along a diagonal; the* B^* *variant then needs* $\omega s^* \le \delta_{in}$*, the inscribed cone angle.* [proved; not machine-checked] [Lean-checked: Theorems B and B^* (incl. upright) for proper-rotation attitude paths with total body rate $\le \omega$, free roll angle, cone gimbal]

Theorem 3.20 (Independent transverse torque)**.** *In M3a/M4-dyn (axisymmetric* J*, transverse torque* $\|\tau_\perp\| \le \tau_t$ *independent of the engine, any roll,* $\Omega(0) = 0$*), every body axis satisfies the envelope* $\angle(b(s), b(s')) \le \Psi(s') - \Psi(s)$ *with* $\Psi(s) = \int_0^s \min(\bar{\alpha} r, \omega)\, dr$*,* $\bar{\alpha} = \tau_t / J_t$*, and the radial "bang" arcs attain it. Hence (i) with terminal attitude free and* $\delta + \Psi(t_f) \le \pi/2$*,* F_{t_f} *is nonconvex; (ii) with terminal attitude* $b(t_f) = e_z$*,* $\Omega_\perp(t_f) = 0$ *and unlimited gimbal rate, a dive-then-brake face with a sufficiently short dive* s^* *(*$\bar{\alpha} s^* \le \omega$*,* $\bar{\alpha} s^{*2} \le \delta_{in}$*,* $4s^* \le t_f$*,* $\delta^* + \bar{\alpha} s^{*2}/2 \le \pi/2$*) again gives nonconvexity, with altitude and glide-slope constraints handled as in Theorem* B^**.* [proved; not machine-checked] *(the state-constraint clause of (ii) is inherited from Theorem* B^* *and was not independently re-derived)*

The gyroscopic terms do no work on $\|\Omega_\perp\|$ for axisymmetric J, so roll and coning cannot defeat the envelope. Not covered are a nonzero initial transverse rate (which breaks the $\pm u$ symmetry), nonaxisymmetric inertia, and gimbal-rate limits in the terminal-attitude case.

3.8.1 Pure TVC: what is proved

On a pure-TVC vehicle the M1 witnesses are not realizable: deflecting the thrust outward rotates the body inward. The centre-of-percussion identity (Lemma 2.1) converts the problem into a zero-gimbal, angular-acceleration-limited problem for $y = r + \varepsilon b$, with boundary data that map affinely. The landing condition becomes a fixed affine condition on $(y, \dot{y})$ only when the vehicle lands upright at rest, $b(t_f) = e_z$, $\Omega_\perp(t_f) = 0$.

Theorem 3.21 (Pure TVC, rest-terminal rate, tilt free)**.** *Model PT with* constant mass *(so* ε *and* L/J_t *are constant), zero roll rate imposed as a model restriction, axisymmetric* J*, unlimited gimbal rate (cone or box gimbal),* $b(0) = e_z$*,* $\Omega_\perp(0) = 0$*, and landing* $r(t_f) = v(t_f) = 0$*,* $\Omega_\perp(t_f) = 0$ *with terminal tilt* free*. In the regime (H-f)* $\varepsilon\bar{\omega}^2 < \rho_1 \cos\delta^*/m_0$ *(strict) and (H-*π*)* $\bar{\omega} t_f \le \pi/2$*, for a rate cap* $\bar{\omega}$*,* $\operatorname{cl} F_{t_f}$*, and hence* F_{t_f}*, is nonconvex, with no state constraints (proof sketch with the closure step in Appendix D).* [proved; not machine-checked]

Theorem 3.21 does not cover the realistic case: an upright terminal attitude with an altitude constraint. That case is settled at four horizons by computer-assisted theorems certified in exact arithmetic: $t_f = 1$ and 2 s without any tilt cap, and $t_f = 3$ and 5 s in the tilt-capped class.

Theorem 3.22 (Theorems TVC-1, TVC-2, TVC-3, TVC-5: pure TVC, upright at rest, $r_z \geq 0$; computer-assisted)**.** *Model PT (pitch and yaw torque produced only by the gimbal), with constant mass and constant $\varepsilon = J_t/(mL)$, axisymmetric inertia with the gimbal pivot on the body axis (the vehicle's centimetre-scale pivot offset and products of inertia are neglected), and* unlimited gimbal rate. *Use the GTPL wet constants ($m = 79.88$ kg, $\rho_1 = 400$ N, $\rho_2 = 1000$ N, $J_t = 63.6$ kg m^2, $L = 0.589$ m, $\delta = 15°$). Start from $b(0) = e_z$, $\Omega(0) = 0$; land at rest and upright, $r(t_f) = v(t_f) = 0$, $b(t_f) = e_z$, $\Omega_\perp(t_f) = 0$; and impose $r_z \geq 0$ along the trajectory. Then, on the dive-then-brake face with switch time s^*, an explicit witness pair $\theta_w, R_\pi\theta_w$ lies in the constrained feasible set while its midpoint θ_m does not, so that set is nonconvex. This is certified in the following cases (margin = witness gain minus the bound on every zero-lateral competitor, in metres of ℓ):*

	t_f, s^*	*class*	*gimbal / competitors*	*margin* $G_w - U_{\text{tot}}$	*distance* $\geq$
TVC-1	*1 s, 0.25 s*	*uncapped, $r_z \geq 0$ + glide slope*	*cone 15°, roll held*	9.96×10^{-6} *m (13.7%)*	9.63×10^{-6} *m*
TVC-B	*1 s, 0.25 s*	*same*	*Cardan box ±15°, roll held*	7.57×10^{-6} *m (5.4%)*	7.34×10^{-6} *m*
TVC-1R	*1 s, 0.25 s*	*same*	*cone, free roll via RCS*	2.90×10^{-6} *m (4.0%)*	2.81×10^{-6} *m*
TVC-2	*2 s, 0.5 s*	*uncapped, $r_z \geq 0$ + glide slope*	*cone, roll held*	1.09×10^{-4} *m (8.06%)*	9.2×10^{-5} *m*
TVC-3	*3 s, 0.5 s*	*tilt cap 80° or 30°, $r_z \geq 0$*	*cone, roll held*	5.65×10^{-5} *m (4.16%)*	4.78×10^{-5} *m*
TVC-5	*5 s, 0.5 s*	*tilt cap 80° or 30°, $r_z \geq 0$*	*cone, roll held*	2.7×10^{-6} *m (0.20%)*	2.3×10^{-6} *m*

Here "distance" is $\operatorname{dist}(\theta_m, F_{t_f})$ *to the unconstrained (uncapped rows) or capped (capped rows) feasible set. The glide slope is part of the claim for TVC-1 and TVC-2, with* $\tan\gamma_{gs}$ *up to the witnesses' minimum glide ratio, but not for TVC-3 and TVC-5. The capped rows concern a different problem class, $b_z \geq \cos\theta_{\max}$ imposed on every trajectory. Nonconvexity of the capped feasible set neither implies nor is implied by nonconvexity of the uncapped one. By Theorem 3.1, no uniform convex relaxation with affine data entry is lossless for the corresponding pure-TVC landing problem at these horizons.* [Lean-checked: logical skeleton only: `TVC.tvc1_main` / `tvc1_model` take the certificate and the analytic reduction as explicit hypotheses] *(The margins are certified in exact arithmetic, except that the localization step functions and the physical constants enter as floating-point inputs, independently validated with margins many orders of magnitude above rounding error. The distance bounds were recomputed independently but evaluated in floating point.)* [proved, computer-assisted]

Method. Using the centre-of-percussion identity, the face functional $\ell = r_{0z} + s^* v_{0z}$ is written exactly in terms of the horizontal tilt $h = b_h$. Pointwise inequalities then give a *quadratic majorant* of the gain of *every* competitor with zero lateral data, which is exactly the condition θ_m satisfies. The torque bound enters through an S-procedure multiplier $\lambda(s)$, and the two lateral-moment conditions through Finsler multipliers. The quadratic form is a sum over the two components of h, so the vector bound equals the scalar bound. Planarity and rigidity are never used, and coning, loops and two-lobe paths are covered automatically. For $t_f = 1$ s the resulting infinite-dimensional inequality is reduced to a $2N \times 2N$ SDP ($N = 60$ or 120). The reduction uses a piecewise-linear cell basis, with Wirtinger-type cell-local remainders and a localization lemma that shrinks the a priori speed and tilt bounds for competitors with nonnegative gain. For $t_f = 2, 3, 5$ s the same quadratic form is certified negative by a backward Riccati (dynamic-programming) recursion over the cells, which is $O(N)$ and handles up to 1500 cells. The bootstrap localization is iterated to a fixed point. The multipliers are generated by coarse SDPs, which are not part of the proof; each final multiplier

is checked by the recursion. The witness is an explicit ordinary control, a fast $\pm\delta$ bang-bang realization with exact terminal correction, whose gain G_w is evaluated in interval (Arb) arithmetic. For $t_f = 1$ s an independent re-implementation rebuilt every certificate matrix by exact polynomial integration (rationals and Arb balls) and decided definiteness by exact integer Sylvester minors. For $t_f = 2, 3, 5$ s it re-ran each recursion independently in Arb (256 bits) with exactly integrated stage data, certifying every pivot; the error radii are about 10^7 times below the thinnest margin. A separate attempt to break TVC-2 failed. For $t_f = 1$ s three repairs were made, and all were carried out and verified: (W1) the originally stored witness was a relaxed (chattering) control and was replaced by the explicit bang-bang one; (W2) the gyroscopic weight in the free-roll case was replaced by a rigorous upper weight; (W3) the cell-local remainder constants now use the true supremum rather than a maximum over quadrature nodes. □

Scope, honestly.

- *The gap is thin.* The certified distances are 10^{-6}–10^{-4} m, for the same reason as Remark 3.9. TVC-5's margin is only 0.20%, but it is rigorous.
- *Unlimited gimbal rate is essential to the certificate.* The bang-bang witness needs sub-millisecond gimbal reversals. Against a 100°/s gimbal-rate limit, the deflection credit in the unlimited-rate baseline is unattainable.
- *Constant mass only.* Mass depletion makes ε time-varying and moves the baseline by more than the gap.
- *Horizons and classes.* Without a tilt cap, $t_f = 1$ and 2 s are certified. With a tilt cap, $t_f = 3$ and 5 s are certified. The cone gimbal with roll held is certified at every one of these horizons, but the box gimbal and free-roll competitors only at $t_f = 1$ s. At $t_f = 7$ s (capped) the certificate fails by 4.8%.

The fully realistic model, numerically. The model here combines a ±15° Cardan box gimbal rate-limited to 100°/s from centred, mass depletion, free roll, $r_z \geq 0$ and an upright-at-rest touchdown. In it, the zero-lateral maximum of the dive-then-brake face lies strictly below the free maximum at every horizon tried:

$t_f = 3$ s, $s^* = 0.75$ s	$t_f = 5$ s, $s^* = 0.5$ s	$t_f = 5$ s, $s^* = 1$ s	$t_f = 7$ s, $s^* = 1$ s
1.26×10^{-2} m	4.8×10^{-3} m	2.8×10^{-2} m	3.3×10^{-2} m

These are local solutions of discretized NLPs, not certificates. The status of the realistic model is therefore *nonconvex numerically, unproved.* [numerical evidence only]

Open.

- *Longer horizons.* Open: $t_f = 7$ s in the capped class (fails by 4.8%); uncapped $t_f > 2.3$ s, which needs a lemma excluding tumbling competitors, because the a priori tilt bound no longer keeps the body above the horizontal; and the box gimbal and free-roll competitors beyond $t_f = 1$ s.
- *A rate-limited certificate,* adding the jerk constraint implied by the gimbal-rate limit with its own S-procedure multiplier.
- *Mass depletion; free final time* (covered only conditionally, via Lemma 3.10).

Other evidence for pure TVC: a torque-coupled hop instance has a 1.74 m hole in its landing set (Section 5), and with $\rho_1 = 0$ pure TVC is nonconvex in a strong-torque, small-gimbal asymptotic regime .

Remark 3.23 (Consistency with 3-DOF). For the exact 3-DOF problem with free direction, F_{t_f} is convex even with mass depletion, and with any measurable compact control set it is convex for constant mass, by Aumann's theorem [proof sketch only]. Theorem A therefore never contradicts 3-DOF LCvx in any case where convexity can be decided. The decisive ingredient of our obstruction is that the attitude constraint *couples different times*, through continuity and a rate bound. Pointwise (measurable) attitude constraints never obstruct.

4 What is possible

The impossibility results identify the obstruction precisely: the *choice* of a rate-limited attitude profile. Once that choice is removed, convexification works again, and in the pure-TVC case it works without any relaxation. This section collects the positive results. They are of two kinds: exact statements once the attitude profile is fixed (Section 4.1 to Theorem 4.4), and a single convex program for the pure-TVC body that is sound without being complete (Section 4.5). Aumann's theorem [28] is the one external theorem used without proof. The Pontryagin maximum principle [33, 35] enters only as a labelled hypothesis (H-PMP).

4.1 LCvx is exact for any fixed attitude profile

Fix a measurable $b : [0, t_f] \to S^2$. The original problem $\mathrm{P}_0(b;\theta)$ minimizes $\int \|T\|$ subject to $\rho_1 \le \|T\| \le \rho_2$, $\langle b, T\rangle \ge \|T\| \cos\delta$, the dynamics (1) and the boundary conditions. The relaxation $\mathrm{P}_1(b;\theta)$ minimizes $\int \Gamma$ over $T \in K_\Gamma(b) := \{\|T\| \le \Gamma,\ \langle b, T\rangle \ge \Gamma\cos\delta\}$, $\rho_1 \le \Gamma \le \rho_2$. This is the pointing-cone LCvx of [4] with a time-varying axis.

Theorem 4.1 (Theorem C-W: value and feasibility exactness for any b)**.** *For every* P_1*-feasible* (T, Γ) *there is a measurable* $\tilde{T}(t) \in \{\|\tilde{T}\| = \Gamma,\ \angle(\tilde{T}, b) \le \delta\}$ *with the same* Γ*, mass history and cost, whose state agrees with that of* (T, Γ) *at* t_f *and at any prescribed finitely many times. Hence* P_0 *and* P_1 *have the same feasible data and the same value, attainment transfers, and nodewise state constraints are preserved. With constant mass (or the convexified-mass model),* $F_{t_f}(b) = \{\theta : \mathrm{P}_0(b;\theta)$ *feasible*$\}$ *is convex (Corollary C-F).* [proved; not machine-checked] [Lean-checked: feasibility part and C-F, constant mass, no state constraints, Aumann as explicit hypothesis]

Proof idea. On each subinterval the state increment is linear in T/m for the fixed mass history. The extreme points of the cap $K_\Gamma(b)$ are exactly its spherical part. Aumann's theorem replaces the relaxed selection by a lossless one with the same moments. □

With convex path constraints the value statement holds under a uniform Slater condition (a relaxed-feasible trajectory at positive distance from the constraint boundary at *all* times). Attainment of P_0 is not claimed in that case. *Caveat:* such a Slater point cannot exist when a path constraint is active at the fixed touchdown state, as for the glide-slope cone with apex at the target or for $r_z \ge 0$ with $r_z(t_f) = 0$, i.e. for the standard constraint set (3). For that set the value statement with path constraints is therefore not established; the nodewise statement is.

Theorem 4.2 (Theorem C-S: every relaxed optimum is lossless, under PMP)**.** *Assume (H-PMP) for a* P_1*-optimal solution, and let* $p = a + ct$ *be the (affine, for every* b*) primer vector. Off* $D = \{t : p(t) \in -\mathbb{R}_{\ge 0} b(t)\}$ *the optimum is lossless and pointwise unique. With constant mass the lossy set is contained in* $D \cap \{\Gamma^* = \rho_1\}$*. D has positive measure only if* $p \equiv 0$*, or b dwells at* $\pm$ *a fixed vector, or b follows a* gnomonic line $\pm(a + ct)/\|a + ct\|$ *on a set of positive measure. For analytic b,* $\det(b, \dot{b}, \ddot{b}) \not\equiv 0$ *suffices for nondegeneracy. Degenerate profiles form a meagre subset of the*

complete metric space of admissible *attitude profiles (*(AL_ω) *with the attitude boundary conditions, sup metric).* [proved; not machine-checked] [Lean-checked: under `PMPConclusion`; constant-mass parts; analytic and meagre criteria not formalized]

The meagreness claim must be made in the admissible class, not in $C^0([0,t_f];S^2)$, where the admissible class is itself nowhere dense. The proof runs in that class, because piecewise constant-speed geodesic interpolants (with short out-and-back arcs where needed) are dense in it and are nondegenerate. With velocity constraints (e.g. a speed limit) the adjoint λ_v can carry a measure part. The pointwise statement of Theorem 4.2 still holds a.e., but the affine primer and the gnomonic structure apply only on open intervals where every state constraint is inactive. The result is sharp. For a fixed upright cone and a vertical descent at a suitable altitude, an interior-point solver returns a lossy relaxed optimum with all lossy nodes at minimum throttle. A one-step "purification" SOCP repairs it (Claims C-X, C-K; discrete version: at most three lossy nodes without state constraints, Lean `purification_three`). So *the relaxation itself is never the problem.* This agrees with the lossiness observed in stored GTPL solutions (296 N delivered on a 400 N minimum-throttle arc).

4.2 Per-instance exactness and the union structure

Theorem 4.3 (Theorem C-U, no path constraints)**.** *Without path (state) constraints,* $F_{t_f} = \bigcup_b F_{t_f}(b)$ *with each* $F_{t_f}(b)$ *convex (M1 with constant or convexified mass; PT by Theorem 4.4), and* $V(\theta) = \min_b V(b;\theta)$*. In M1 with constant mass the 5-DOF minimum is attained, and the convex program at an optimal attitude* b^* *is exact for that instance.* [proved; not machine-checked]

The impossibility theorems say exactly that this union of convex sets is not convex. *All* of the nonconvexity lies in the map $\theta \mapsto b^*(\theta)$.

4.3 Pure TVC: exactly convex given the attitude

Theorem 4.4 (Theorem C-6)**.** *In model PT with zero roll rate and constant mass, Euler's equation is equivalent to* $\Pi_b T = -k\Pi_b \ddot{b}$*,* $k = J_\perp/L$*,* $\Pi_b = I - bb^\top$*. For a fixed* $b \in W^{2,\infty}$*, write* $F = -k\Pi_b\ddot{b}$*,* $\varkappa = \|F\|$*. The exact 6-DOF problem is the scalar-throttle program*

$$\min \int \sqrt{T_a^2 + \varkappa^2} \quad s.t. \quad \ddot{r} = (T_a b + F)/m + g, \quad \max\{\varkappa \cot\delta, \sqrt{(\rho_1^2 - \varkappa^2)_+}\} \le T_a \le \sqrt{\rho_2^2 - \varkappa^2},$$

which is convex. No relaxation is involved. Consequently $F_{t_f}^{PT}(b)$ *is convex for every attitude profile. If the body is held upright (*$b \equiv e_z$*), then* $T \parallel e_z$*: the effective cone is* 0*, so the common "upright body with a* 15° *pointing cone" 3-DOF model is not 6-DOF-feasible for any off-vertical thrust.* [Lean-checked: constant mass; torque balance and convexity joined in prose]

4.4 Exact subclasses of the uniform problem

Proposition 4.5 (X5: small pointing cone, M1 only)**.** *In M1 with an LCvx pointing constraint* $\angle(d, e_z) \le \theta_p$*: if* $\theta_p \le \delta$ *and the initial (and, if prescribed, terminal) attitude lie within* $\delta - \theta_p$ *of* e_z*, and, if the terminal attitude* b_f *is prescribed,* $\omega t_f \ge \angle(b_0, b_f)$ *(automatic for* $b_0 = b_f = e_z$*), then* $F_{5\mathrm{DOF}} = F_{3\mathrm{DOF}}(\theta_p)$*, so classical LCvx guarantees transfer verbatim. If* $\theta_p > \delta$ *(and the* B^* *braking condition holds),* F_{t_f} *is nonconvex. Thus* $\theta_p = \delta$ *is a sharp threshold* for M1. [proved; not machine-checked]

X5 relies on the gimbal pointing the thrust while the body stays still, which is impossible on a pure-TVC vehicle (Theorem 4.4). There the effective exact class collapses to $\theta_p = 0$. More

precisely, a prescribed planar thrust history with $\|T\| \geq \rho_1$, an initial axis b_0 in the plane and an in-plane initial body rate, is exactly TVC-realizable only for rates in an interval of length at most $2\delta\mu_c/\sinh(\mu_c t_f)$, $\mu_c^2 = L\rho_1 \cos\delta/J_\perp$. For our vehicle at $t_f = 7$ s this is 3.5×10^{-6} rad/s at $\rho_1 = 400$ N (1.8×10^{-5} rad/s at the flight-code value $\rho_1 = 300$ N). So a 3-DOF LCvx optimum is essentially never exactly 6-DOF-optimal (Claim C-Y, [proved; not machine-checked]). In M1, by contrast, $V_5 = V_3$ iff some 3-DOF optimum is δ-trackable by an admissible attitude, with a closed-form primer-arc certificate (Claim C-3D) [proof sketch only].

4.5 A sound single convex program for pure TVC

Theorem 3.1 excludes a convex program that is *sound* (feasible only at data from which a landing exists) and *complete* (feasible at all such data). It does not exclude a program that is only sound. Proposition 4.5 is one, but its feasible set is small: the body never has to turn. For the pure-TVC model the centre-of-percussion identity gives a much larger one, because in the coordinates $y = r + \varepsilon b$ of Lemma 2.1 the dynamics are linear and the attitude is not a separate unknown. This is differential flatness [22]. For the planar VTOL aircraft the flat output is the same point, the Huygens centre of oscillation [23]; flat outputs underlie minimum-snap planning for multirotors [24] and have been used for planetary-landing guidance [25]; and planning a flat output as a B-spline, with constraints imposed on control points so that they hold between nodes, is established for differentially flat systems [26]. B-spline parametrizations also appear in sequential convex programming for powered descent [36]. We did not find the combination used here: the centre-of-percussion output of the gimballed rigid body, with the minimum-throttle and gimbal limits imposed as convex restrictions, giving one SOCP with a soundness guarantee.

Setting. Model PT with zero roll rate and mass depletion $\dot m = -\alpha\|T\|$, thrust $T = T_a b + T_\perp$ ($T_\perp \perp b$), limits $\rho_1 \leq \|T\| \leq \rho_2$, gimbal angle $\mu \leq \delta$ ($\tan\mu = \|T_\perp\|/T_a$) and tilt $\angle(b, e_z) \leq \vartheta_{\max} < \pi/2$.

Assumption 4.6 (Fixed centre of percussion)**.** $\varepsilon = J_t/(mL)$ is constant along the flight.

With constant mass this is automatic. With depleting mass it says that J_t scales as mL; for the motivating vehicle the simulator's mass table gives $\varepsilon = 1.352$, 1.340 and 1.321 m at 79.9, 75 and 70.4 kg.

The flat map. For $y \in C^4$ put $u = \ddot y - g$ and $f = \|u\|$. Wherever $u \neq 0$ define

$$
\begin{gathered}
b = \frac{u}{f}, \qquad \dot b = \frac{\Pi_b \dot u}{f}, \qquad \Pi_b \ddot b = \frac{\Pi_b(\ddot u - 2(\dot f/f)\dot u)}{f}, \quad \dot f = \langle b, \dot u\rangle, \\
a = f + \varepsilon\|\dot b\|^2, \qquad \ell = -\varepsilon\,\Pi_b \ddot b, \qquad \tau = \sqrt{a^2 + \|\ell\|^2}, \\
m(t) = m_0 \exp\Big(-\alpha \int_0^t \tau\Big), \qquad T_a = ma, \qquad T_\perp = m\ell, \qquad r = y - \varepsilon b.
\end{gathered} \tag{6}
$$

So the whole state and both controls are algebraic in y and its first four derivatives, and the mass is a quadrature.

The program. The unknown is y, a clamped quintic B-spline with S uniform spans of length $h = t_f/S$. On a span, $u, \dot u, \ddot u$ are polynomials of degree $3, 2, 1$. Raising their Bézier control points to

degree 3 gives four *control tuples* (U^j, U_1^j, U_2^j), linear in the spline coefficients, with

$$(u,\dot u,\ddot u)(t)=\sum_{j=0}^{3} B_j(t)\,(U^j,U_1^j,U_2^j),\qquad B_j\ge 0,\quad \sum_j B_j=1,\quad \int_{\text{span}} B_j=\frac{h}{4}. \tag{7}$$

The data are r_0, v_0, the current body axis b_0, its rate $\dot b_0$, the transverse angular acceleration β_0 that the gimbal's current deflection produces, and m_0. On each span fix four numbers $\eta,\kappa\in\mathbb{R}$, $\phi>0$, $\ell_b>0$ (the *references*; any values are allowed). With $p=\langle U,e_z\rangle$, $f_0=\langle b_0,u(0)\rangle$ and auxiliary scalars e_1,q,k,g_2,s_f,σ per tuple, the program is

$$\begin{aligned}
&\min\ \tfrac{h}{4}\textstyle\sum\sigma\\
&\text{s.t.}\ \ y(0)=r_0+\varepsilon b_0,\quad \dot y(0)=v_0+\varepsilon\dot b_0,\quad u(0)=f_0b_0,\quad \Pi_{b_0}\dot u(0)=f_0\dot b_0,\\
&\qquad \Pi_{b_0}\ddot u(0)=f_0\beta_0+2\langle b_0,\dot u(0)\rangle\dot b_0,\qquad y(t_f)=\varepsilon e_z,\quad \dot y(t_f)=0,\quad u_h=\dot u_h=\ddot u_h=0\ \text{at}\ t_f,\\
&\text{(R1)}\ \ \|U_h\|\le\tan\vartheta_{\max}\,p,\qquad \text{(R2)}\ \ p\ge\ell_b,\\
&\text{(R3)}\ \ \|U_1-\eta U\|\le e_1,\quad e_1^2\le q\,p,\\
&\text{(R4)}\ \ \|U_2-2\eta U_1-\kappa U-ke_z\|+\sin\vartheta_{\max}\,|k|\le g_2,\qquad G:=g_2+q,\\
&\text{(R5)}\ \ \varepsilon\,G\le\tan\delta\,(2\phi p-\phi^2),\\
&\text{(R6)}\ \ \|U\|\le s_f,\quad \big\|(s_f+\varepsilon q/\ell_b,\ \varepsilon G/\ell_b)\big\|\le\sigma,\\
&\text{(R7)}\ \ p\ge\rho_1\,c_s(z_{lo,s}),\quad \sigma\le\rho_2\,\tau_s(z_{up,s}),\quad z_{lo,s}\in[z_s^{\min},z_s^{\max}],\quad z_{lo,S}\ge\ln m_{dry}.
\end{aligned} \tag{8}$$

Here $z_{up,s}=\ln m_0-\alpha(\dot y_z(t_s)-\dot y_z(0)+g_0t_s)$ at the start t_s of span s and $z_{lo,s}=\ln m_0-\alpha\frac{h}{4}\sum_{\text{spans}\le s}\sum_j\sigma^j$ are affine in the unknowns, c_s is the chord of e^{-z} on a fixed interval $[z_s^{\min},z_s^{\max}]$ and τ_s is a tangent of e^{-z}. Every row is linear or second-order-cone, so (8) is an SOCP; the boundary rows are linear because f_0 and $\langle b_0,\dot u(0)\rangle$ are linear in the coefficients. (r_0,v_0) enter affinely, exactly as in LCvx.

Theorem 4.7 (Theorem F: soundness)**.** *Under Assumption 4.6, let y be feasible for (8), with $\angle(b_0,e_z)\le\vartheta_{\max}$. Then model PT, started from $(r_0,v_0,b_0,\dot b_0,m_0)$ and driven by the thrust $T=T_ab+T_\perp$ of (6), has the unique solution $r=y-\varepsilon b$ with body axis b and mass m, and this solution*

(i) starts with the gimbal where it is: $T_\perp(0)=-m_0\varepsilon\beta_0$;

(ii) lands at rest, upright, with zero body rate and a centred gimbal: $r=\dot r=0$, $b=e_z$, $\dot b=0$, $T_\perp=0$ at t_f;

(iii) satisfies $\rho_1\le\|T\|\le\rho_2$, $\mu\le\delta$ and $\angle(b,e_z)\le\vartheta_{\max}$ at every *$t\in[0,t_f]$;*

(iv) burns at most $m_0(1-\exp(-\alpha\frac{h}{4}\sum\sigma))$ and ends above m_{dry}.

[proved in Appendix E; not independently refereed, not machine-checked]

The proof is in Appendix E. Its steps: Lemma 2.1 removes the gimbal force; the flat map is the unique solution (Lemma E.1); the boundary rows are equivalent to the stated initial and terminal conditions (Lemma E.2); five pointwise bounds turn (R2)–(R6) into the actuator limits (Lemma E.3); and by (7) a constraint that is jointly convex in $(u,\dot u,\ddot u)$ and imposed on the control tuples holds at every instant (Lemma E.4). No relaxation is used, so there is no tightness to establish.

Proposition 4.8 (The price: incompleteness)**.** *For fixed $b_0,\dot b_0,\beta_0,m_0,t_f$ and references, the set of (r_0,v_0) at which (8) is feasible is convex and is contained in the feasible boundary-data set of the model with these limits. Whenever the latter is not convex, the containment is strict.* *[proved in Appendix E; not independently refereed, not machine-checked]*

The feasible set of the pure-TVC model is nonconvex in the cases of Theorem 3.22, and numerically at the planning limits used in Section 6, so (8) must refuse some landings that exist. The loss has three sources. Rows (R2) and (R7) bound the thrust from below by its vertical component, so the program never plans to shed thrust by tilting, which is the manoeuvre the obstruction of Theorem 3.7 is built from. Row (R4) bounds the interaction of throttle changes with rotation instead of computing it. And control polygons contain the curves they bound. Section 6 measures the size.

Remark 4.9 (References). The numbers $\eta, \kappa, \phi, \ell_b$ change how much of the feasible set (8) keeps, never whether its points are flyable: Theorem 4.7 holds for any values. The restriction is tightest at $\eta = \dot{f}/f$, $\kappa = (\ddot{f} - 2\eta\dot{f})/f - \|\dot{b}\|^2$, $\phi = p$. The experiments of Section 6 use the fixed values $\eta = \kappa = 0$, $\phi = 7.5\ \mathrm{m/s^2}$, $\ell_b = \rho_1/m_0$ unless stated.

Remark 4.10 (What the theorem does not cover). Gimbal and throttle slew rates depend on $y^{(5)}$ and are not constrained by (8); in the experiments they are evaluated exactly on every plan and a plan that violates them is not flown. Altitude and glide-slope constraints on r are imposed at sampled instants only. Assumption 4.6 is a modelling assumption. The theorem is proved on paper and tested numerically (458 feasible random plans, each evaluated at 1201 instants, no violation; open-loop replays through the rigid body within 23 nm); it is not formalized.

4.6 Semidefinite lifts

Occupation-measure linear programs with point-mass data entry are exact but infinite-dimensional and per instance (Section 2.5). Their order-one truncation is exactly the affine case of Theorem A. The first nontrivial tractable truncation is the degree-2 (Shor) lift [37, 38]. We study it for the fixed-thrust minimum-landing-error problem. With weights c_k^v, c_k^r, unknown unit vectors b_k (axis) and d_k (thrust direction), inner-product constraints $\langle d_k, b_k\rangle, \langle d_k, b_{k+1}\rangle \geq \cos\delta$ (gimbal) and $\langle b_k, b_{k+1}\rangle \geq \cos\omega h$ (rate), and cost $\|\sum c_k^v d_k - w_v\|^2 + \|\sum c_k^r d_k - w_r\|^2$, the problem is a vector QCQP. The data w enter only the objective, and the lifted cost is a second-moment cost. The lift therefore lies outside the classes of Theorems A and D, and no contradiction arises.

Lemma 4.11 (Shor lift = Hilbert-sphere problem)**.** *The Shor relaxation's value equals the minimum of the same vector QCQP posed in $\mathbb{R}^{3+m}$ (indeed in any real Hilbert space containing $\mathbb{R}^3$). The lift is exact iff some optimum can be "planarized" into $\mathbb{R}^3$.* [proved; not machine-checked]

Theorem 4.12 (Stress certificate and local-to-global)**.** *Let z^* be a genuine KKT point with multipliers (λ, μ) and stress matrix $Q_\lambda = C + \mathrm{diag}(\mu) - \frac{1}{2}\sum_e \lambda_e(E_{ij} + E_{ji})$. If $Q_\lambda \succeq 0$, then z^* is a global optimum of the lift in every dimension, hence of the genuine problem. Conversely, exactness implies that a PSD stress exists, and $Q_\lambda \succ 0$ gives a unique rank-3 optimum. Moreover the lift is exact iff some genuine global minimizer is a* local *minimizer of the problem posed in $\mathbb{R}^4$ ("$\Leftarrow$" under LICQ). Any LICQ KKT point that is a local minimizer in $\mathbb{R}^4$ is a global minimizer of the lift.* [proved; not machine-checked]

Theorem 4.13 (Coplanar data: certified global optima)**.** *If the lateral parts of w_v and w_r are parallel (velocity-only, position-only or vertical-plane targets), then every planar genuine local minimizer satisfying LICQ is a certified global optimum of both the lift and the genuine problem. This remains true with additional constraints that are linear in the Gram matrix, e.g. altitude $\langle r_j, e_z\rangle \geq 0$ at the nodes.* [proved; not machine-checked]

A glide-slope constraint is not linear in the Gram matrix (it involves squares of first moments), and the proof does not extend to it. With a glide slope, exactness must be checked a posteriori by

the stress test of Theorem 4.12; an a priori statement is open. A structural fact about lifted optima also holds: under rate saturation with nondegenerate multipliers, every lifted KKT configuration lies in $\mathrm{span}(e_z, b_1, y_v, y_r)$, and since the target subspace $E_w = \mathrm{span}(e_z, w_v, w_r)$ is contained in that span, the lifted optimum has rank ≤ 4 when the target data span three dimensions (≤ 5 for coplanar data) [proved; not machine-checked].

Where exactness fails (all [numerical evidence only] unless marked).

- *Unit-thrust discretization with gimbal.* With $\|d_k\| = 1$ and $\delta > 0$ the lift is lossy. On an asymmetric near-dent target the SDP gives 0.027529 (rank 4), against a best genuine value of 0.028231 over about 1,000 diverse runs, a 2.5% gap. Three independent implementations agree, and an explicit feasible $\mathbb{R}^4$ chain attains the SDP value. At small N the gap is certified by order-2 Lasserre bounds, up to solver accuracy (e.g. $N = 3$, $\delta = 15°$: Shor 1.862 against Lasserre-2 2.358 $\approx$ genuine 2.357). A plausible mechanism is identified: the lifted gimbal lens is exactly the convex hull of the genuine lens , so in the lift d_k can tilt into the extra dimension at no rate cost and shorten its in-plane part. A unit-norm discretization forbids this, while relaxed controls in continuous time allow it. With the convexified lens ($\|d_k\| \leq 1$) the same instances are exact numerically.
- *Variable throttle.* Every degree-2 throttle lift tested is lossy. For the natural lift with $\|U_k\|^2 = a_k^2$, the gap at fixed physics grows with the discretization: 12%, 40%, 54%, 63% and 68% at $N = 4, 6, 8, 12, 16$. The lift tends to the no-rate-limit value, so the gap is not a discretization artifact. At $N = 3$ an order-2 Lasserre bound equals the local optimum 0.6047 while the Shor value is 0.5759 [certified numerically by Lasserre-2, up to solver accuracy]. The separate-axis lift CSD3 is also lossy, by 2–35% even with $\delta = 0$.
- *Path constraints.* An altitude constraint stayed exact in every active scenario. A glide-slope constraint broke exactness in 1 of 8 active scenarios (1.19%, rank 4) even with $\delta = 0$, consistent with the restriction of Theorem 4.13 to Gram-linear constraints.
- $\delta = 0$ *(no gimbal), no path constraints.* No gap on about 2,900 targets.

Conjecture 4.14 (CSD2′)**.** *For fixed thrust and no path constraints, the Shor lift of the minimum-landing-error problem is exact in each of three settings: (a) in continuous time, with relaxed thrust direction $u(t) \in \mathrm{conv}\,\mathrm{cap}(b(t), \delta)$ and a rate-limited axis; (b) for the convexified-lens discretization $\|d_k\| \leq 1$; and (c) for the $\delta = 0$ chain.* [open]

No counterexample is known. About 4,500 solves were run, including zero gaps on all 509 instances that were gaps for the unit-norm discretization once convexified. The following partial results have written proofs :

- *T1 (alignment).* The lifted value is an infimum over lifted trajectories whose physical readout lies in $\mathbb{R}^3$. Hence the conjecture for all targets is equivalent to $R_X \cap (\mathbb{R}^3)^2 = R_3$: every lifted trajectory that lands in $\mathbb{R}^3$ can be replaced by a 3-D one with the same terminal state. [proved; not machine-checked]
- *T2 (primer-plane fold).* The lift is exact for *every linear objective of the terminal readout* (V, R), in continuous time and in all three discretizations. This closes the support-function question (it does not cover attitude-dependent linear costs). [proved; not machine-checked]
- *T3–T4 (stress certificate and hemisphere/Picone theorem).* The continuous-time version holds given a normal KKT multiplier set, which is automatic with free terminal attitude but not with prescribed terminal attitude. The discrete version (convexified lens, $\delta = 0$) is fully rigorous, and its proof uses exactly the sign $\mu_d \geq 0$ that the convexified lens provides. An a priori class follows:

targets whose vertical demand exceeds full upright thrust, with $\omega t_f < \pi$, given the multiplier hypothesis. [proved; not machine-checked]

- *T5 (index identity).* At a genuine KKT point satisfying nondegeneracy hypotheses, $\mathrm{neg}(Q) = \mathrm{neg}(Z) - p_0(C_{yw})$. Here Z is the cable part of the stress, and p_0 is the number of nonnegative eigenvalues of the 2×2 lateral residual/target covariance. With free terminal attitude, $\mathrm{neg}(Z)$ equals the number k of crossings of the primer plane, so the lift is exact iff $k \le p_0$. With upright terminal attitude, one residual case $k = p_0 + 1$ remains, and the conjecture reduces to an integer equality there (Conjecture CSD2″). [proved; not machine-checked] *Caveat:* the nondegeneracy hypotheses hold at only a minority of computed optima (188 of 898 in fresh banks; 9 of 17 earlier test cases). The direct stress test of Theorem 4.12 certifies all 17.
- *P6, P7.* Genuine local optima have no free singular arcs, so the lens-hull mechanism of the unit-norm failure has no continuous-time analogue at genuine optima. The unit-norm failure goes with negative norm multipliers $\mu_d < 0$, which the convexified lens excludes (mechanism proved; correlation numerical). [proved; not machine-checked]

Physical meaning of an exact relaxed optimum. The certified 3-D optimum has a genuine rate-limited attitude $b(t)$. Its thrust direction is relaxed, $u(t) \in \mathrm{conv}\,\mathrm{cap}(b(t), \delta)$. By Aumann's theorem there is a measurable full-thrust direction selection in the cap, with the same b, that reaches *exactly* the same terminal state. This selection generally chatters, so it assumes unlimited gimbal bandwidth. It is not a saturated $\pm\delta$ law.

Open problem (CSD2″). What remains of the conjecture is a residual class of genuine optima with prescribed (upright) terminal attitude and exactly $k = p_0 + 1$ crossings of the primer plane. There the index identity leaves a single integer equality undecided. With free terminal attitude, T5 decides exactness under its nondegeneracy hypotheses. Also open is the continuous-time normality step for prescribed terminal attitude, which T3/T4 assume. No proof attempt is currently active.

Practical reading. The degree-2 lift is a polynomial-size SDP whose value is always a rigorous lower bound, and a positive value certifies that exact landing is infeasible. Exactness is certified a posteriori by a rank-3 optimum, by the stress test of Theorem 4.12, a priori in the coplanar case of Theorem 4.13, or (for every linear objective of the terminal readout) by T2. In the tests, it was exact on every realistic in-air target for our vehicle. It is a certificate, not a lossless convexification. It does not handle minimum fuel: fuel is constant for fixed thrust, and the natural variable-throttle lift is lossy.

5 Numerical evidence

Everything in this section is numerical evidence and is not used in any proof.

Method. Discretized nonlinear programs were solved by multistart interior point (CasADi/IPOPT). For M1 the discretization is *inner*: b is piecewise geodesic with node-to-node angle at most ωh, and d_k must lie within δ of both b_k and b_{k+1}. Spherical caps are geodesically convex, so every discrete solution is an admissible continuous-time trajectory, and every reported distance to a reachable set is an *upper* bound on the true one. Multistart cannot certify global optima, and "no gap found" is weak evidence. Models: M1 (planar and 3-D, with coning initial guesses), M1A (angular-acceleration-limited axis), M3 (torque-coupled TVC rigid body, quaternion attitude, gimbal rate 100°/s), and MT (tilt-angle-rate only, azimuth free). Parameters are those of the motivating vehicle: $m_0 = 80$ kg, $T \in [400, 1000]$ N, $\alpha = 5.66 \times 10^{-4}$ s/m, $\delta = 15°$, $J_\perp = 63.6$ kg m^2,

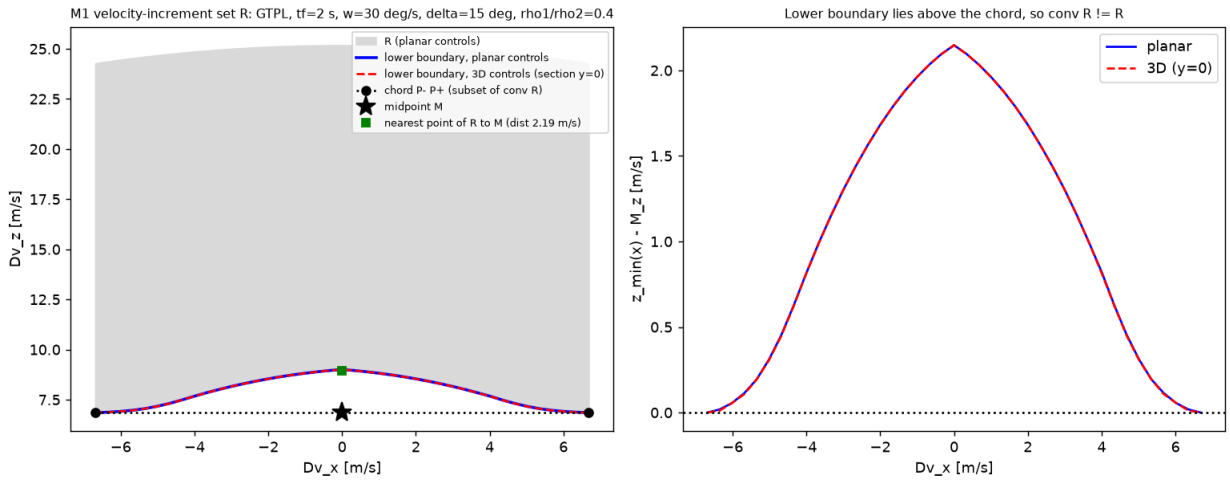


Figure 1: Velocity-increment set R of model M1 ($t_f = 2$ s, $\omega = 30°/\text{s}$, $\delta = 15°$, $\rho_1/\rho_2 = 0.4$). The lower boundary lies strictly above the chord $P_{-u}P_u$, so $M \in \operatorname{conv} R \setminus R$ (Theorem 3.4). Planar and 3-D controls give the same boundary. Numerical (inner discretization).

$L = 0.589$ m, and $\omega = 0.4$ rad/s in the state-constrained studies. A Mars-like parameter set after [2] was also used.

5.1 Theorem B regime: gap law and the ball bound

For $t_f = 2$ s, $\omega = 30°/\text{s}$, $\delta = 15°$, direct searches put R within 2.18–2.23 m/s of the midpoint M, decreasing in N (Fig. 1); an independent search over exactly admissible controls at constant mass reaches 2.14 m/s, against the proved lower bound of 1.81 m/s. 3-D controls gain essentially nothing over planar ones. Over 39 configurations (ω 10–120°/s, ωt_f 15–75°, δ 2–30°, ρ_1/ρ_2 0.05–1, α 0–5×10^{-3}) the empirical law is

$$\operatorname{dist}(M, R)/|P_{u,x}| \;\approx\; 0.31\,\omega t_f,$$

nearly independent of δ, ρ_1/ρ_2 and α, and linear in ρ_1 (Fig. 2). The rigorous ball bound of Theorem 3.5 is 81–83% of the NLP upper bound in all 39 cases, with no violation.

5.2 Beyond the proven regimes (no state constraints)

For long horizons without a tilt cap the generalized midpoint gap grows (2.1/13.5/18.6 m/s at $t_f = 2/4/8$ s), but that regime uses inverted flight. With a 30° tilt cap it is 1.15/1.56/0.35 m/s, shrinking as azimuthal sliding along the cap weakens rigidity. In the tent variant the gaps are 0.66 m/s ($t_f = 2$) and 6.24 m/s ($t_f = 5$). For free final time the Theorem B witness, which starts underground, becomes reachable at a longer horizon (Fig. 3). This is why the free-final-time results need B* and a window. The torque-coupled TVC model M3 has nonconvex R with gaps of 0.8–8 m/s at $t_f = 4$ s, and every minimizer of Δv_z has nonzero lateral component.

5.3 With altitude constraints: the physically relevant sets

Gaps of physically admissible data, all with $r_z(t) \geq 0$ at every node, are summarized in Table 1.

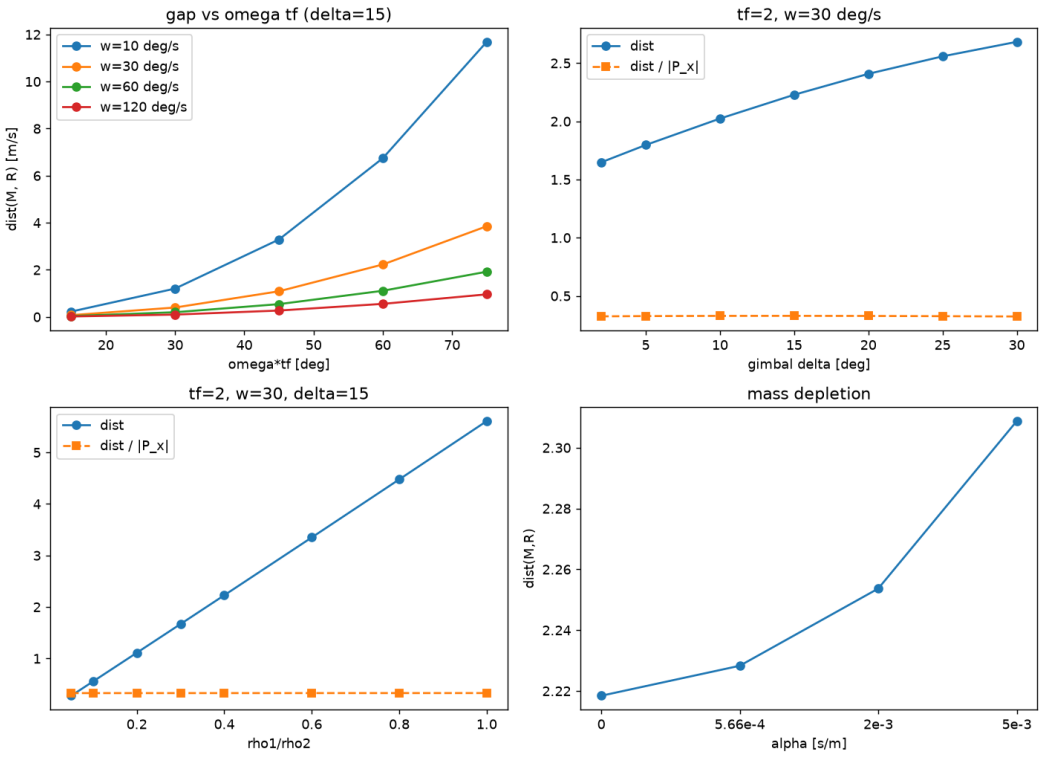


Figure 2: Midpoint gap $\mathrm{dist}(M, R)$ against ωt_f, δ, ρ_1/ρ_2 and α. The normalized gap $\mathrm{dist}/|P_x|$ is flat in δ and ρ_1/ρ_2. Numerical.

- *B^* face.* The rotation-midpoint gap is about 0.010 (weighted m/s) at $t_f = 4$ s and 0.007 at $t_f = 8$ s, stable in N. That is real but tiny, roughly 0.3–0.5% of the face's lateral velocity, consistent with Remark 3.9.
- *Slice lemma (S) at dive-then-brake data.* Strict gaps of 0.016 m ($t_f = 4$) and 0.009 m ($t_f = 8$), with optimizers landing 1.5–2.4 m off axis. Coning narrows planar gaps by at most about 30% (Fig. 4).
- *Face scan and the hop instance.* Among rotation-invariant vertical faces of $F^{\mathcal{X}}$, the pure-dive face is pinned at $r_{0z} = 0$, $v_{0z} > 0$: a vehicle on the pad moving upward. Its midpoint gap is 0.65 ($t_f = 4$) and 2.94 ($t_f = 8$) weighted m/s, about 65 ($t_f = 4$) and 400 ($t_f = 8$) times the B^* gap. At pad data $r_0 = 0$, $v_{0,h} = 0$, $t_f = 4$ s, the landing-offset set L is an annulus whose inner radius grows from 0 to 4.99 m as v_{0z} goes from 7.65 to 7.90 m/s (Fig. 5). Inside the window $v_{0z} \in (7.727, 7.917]$, landing back on the pad is infeasible while landing 0.5–8 m away is feasible. For the torque-coupled TVC model at $v_{0z} = 8.0$ m/s the pad lies 1.74 m from L. The *rigorous* part of the hop result is an exact reduction, an upright pad-landing limit $v_{0z} \le 7.2045$ ($\alpha = 0$) / 7.2833 ($\alpha > 0$), and explicit planar witnesses landing 6.58 m / 4.61 m off the pad at $v_{0z} = 7.7116$ / 7.7425. The *hole* itself is not proved: quadratic and S-procedure certificates are provably too loose here.
- *Tilt-rate-only model MT.* At the same hop data, L is a full disk of radius 10.1 m. The full-direction rate limit, which creates rigidity, is what creates the hole.
- *In-air descent.* From $r_0 = (0, 0, 50)$ m, $v_0 = (4, 0, -5)$ m/s, $t_f = 7$ s, the set L is 104 m wide with

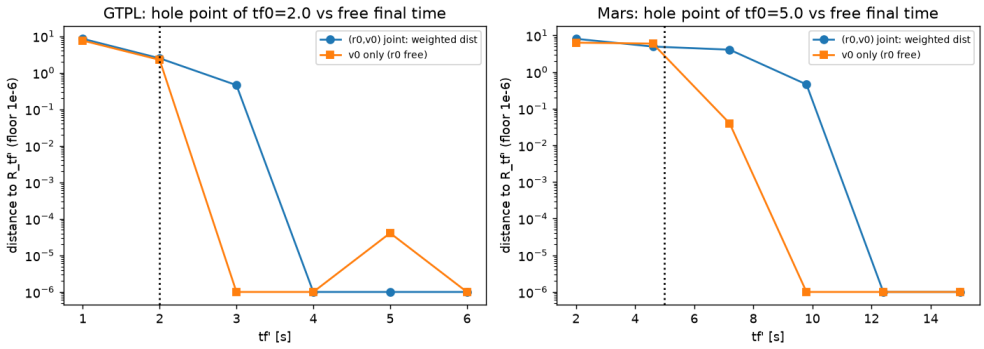


Figure 3: Free final time: the Theorem B hole point at $t_f = 2$ s (an underground datum) is reached at horizons $t'_f \geq 4$ s. The free-final-time obstruction therefore needs the dive-then-brake face and a compact window (Corollary 3.11). Numerical.

Table 1: Size of the nonconvexity at physically admissible data ($r_z \geq 0$), motivating vehicle. All entries numerical (upper bounds on true distances, uncertified global optima) except where marked.

Face / instance	Model	t_f [s]	Gap
B* face, rigorous bound (Theorem 3.7)	M1	7	$\geq 1.1 \times 10^{-5}$ (proved)
B* face, rotation midpoint	M1	4 / 8	≈ 0.010 / 0.007 (weighted m/s)
Slice (S), dive-then-brake	M1	4 / 8	0.016 / 0.009 m in ℓ; offsets 1.5–2.4 m
Pure-dive face (hop, $r_{0z} = 0$)	M1	4 / 8	0.65 / 2.94 (weighted m/s)
Hop landing-set hole	M1	4	up to 4.99 m
Hop landing-set hole	M3 (TVC)	4	1.74 m
Dive-then-brake, upright at rest	PT (6-DOF)	3 / 5 / 7	0.013 / 0.028 / 0.033 m
Same, certified (Theorem 3.22; const. mass)	PT (6-DOF)	1 / 2	$\geq 9.6 \times 10^{-6}$ / 9.2×10^{-5} m (proved)
Same, tilt-capped class, certified	PT (6-DOF)	3 / 5	$\geq 4.8 \times 10^{-5}$ / 2.3×10^{-6} m (proved)
In-air descent (50 m, -5 m/s)	M1	7	none detected (≤ 0.016 m)

no dent above solver noise (≤ 0.016 m).

- *Long horizons and coning.* On the pure-dive face with a 30° tilt cap, the 3-D slice gap falls to 0.69/0.50/0.106 m/s at $t_f = 8/12/20$ s (planar: 0.70/0.80/0.82), and at $t_f = 20$ s the optimizer cones through 359° on the cap. Coning fills about 87% of capped dents, but not all of them.
- *Pure TVC, upright at rest.* Full rigid body, ±15° box gimbal at 100°/s from centred, mass depletion, roll free, $r_z \geq 0$, upright at rest at touchdown, face $\ell = r_{0z} + s^* v_{0z}$: the zero-lateral optimum lies below the free optimum by 1.26×10^{-2} m ($t_f = 3$ s, $s^* = 0.75$ s), 4.8×10^{-3} m ($t_f = 5$ s, $s^* = 0.5$ s), 2.8×10^{-2} m ($t_f = 5$ s, $s^* = 1$ s) and 3.3×10^{-2} m ($t_f = 7$ s, $s^* = 1$ s). The rigorous counterpart is Theorem 3.22 (constant mass, unlimited gimbal rate; $t_f = 1, 2$ s uncapped, $3, 5$ s capped).

5.4 What 3-DOF LCvx asks of the vehicle

The current guidance stack of the motivating vehicle solves a 3-DOF LCvx SOCP and hands only the thrust magnitude to the attitude tracker. Its stored solutions contain thrust-direction jumps of 14.8° within 24 ms and 180° azimuth flips at zero-order-hold switches. These demand roughly 10^3 rad/s^2, against 1–3.5 rad/s^2 available. The relaxation is also not tight on the stored minimum-throttle

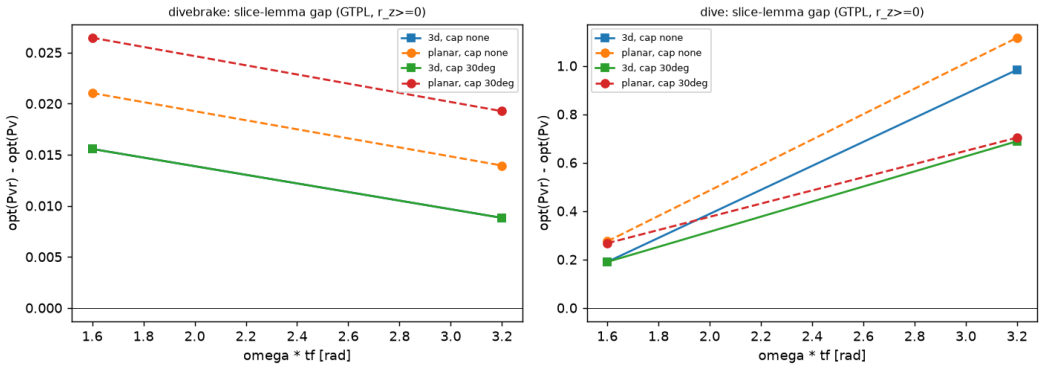


Figure 4: Slice-lemma gaps $\mathrm{opt}(P_{vr}) - \mathrm{opt}(P_v)$ under $r_z \geq 0$ for the dive-then-brake face (left; metres in ℓ) and the pure-dive face (right; m/s), planar versus 3-D (coning), with and without a 30° tilt cap. A strictly positive gap means that the straight-below landing is infeasible while an offset landing is feasible. Numerical.

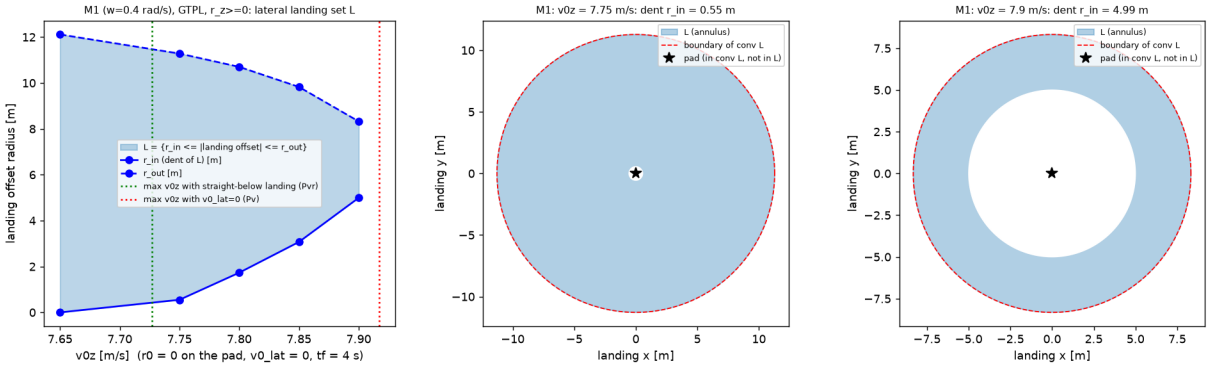


Figure 5: Pad-launched hop ($r_0 = 0$, $v_{0,h} = 0$, $t_f = 4$ s, M1, $r_z \geq 0$): the reachable landing-offset set L is an annulus. The pad lies in $\mathrm{conv}\, L$ but not in L once $v_{0z} \gtrsim 7.73$ m/s. By Theorem D every state-linear relaxation of the minimum-landing-error problem reports error 0 at such data. Numerical; the hole is not certified.

arc (296 N delivered against $\sigma m = 400$ N), consistent with Theorem 4.2, which places lossiness at minimum throttle. Together with Theorem 4.4 (an upright body has effective cone 0), this shows that the 5-DOF/6-DOF coupling is a first-order issue for this vehicle, independent of the size of the nonconvexity.

6 Implications for guidance: a numerical study

This section reports numerical evidence. Its theorem-level ingredients are Theorem 3.1, Theorem 4.3, Theorem 4.4, Theorem 4.7 and the elementary claims listed below. No planner's convergence is proved.

The theory makes four predictions for a real attitude-rate-limited lander.

(P1) *False feasibility.* A uniform convex relaxation that contains every flyable trajectory also accepts boundary data from which no flyable trajectory lands (Theorems 3.1 and 3.7). A 3-DOF guidance program will therefore report "optimal" for some landings the vehicle cannot fly.

(P2) *Where the difficulty lives.* The nonconvexity lies entirely in the choice of attitude profile (Theorem 4.3). Any planner that iterates on the attitude can, in principle, reach the 6-DOF optimum; no single convex program of the class of Theorem 3.1 can do so for all boundary data.

(P3) *Exactness given the attitude.* For a pure-TVC vehicle with a fixed attitude profile, the remaining problem is exactly convex, with no relaxation (Theorem 4.4).

(P4) *Soundness without iteration.* For a pure-TVC vehicle one SOCP in centre-of-percussion coordinates returns only flyable plans (Theorem 4.7), and by Proposition 4.8 it must give up some fuel and some feasible landings in exchange.

We test these on the motivating vehicle (80 kg, 400–1000 N, $\pm 15^\circ$ box gimbal at 100°/s, inertia and lever arm from the club simulator's mass table) with five planners .

6.1 Planners compared

- **A**: a faithful re-implementation of the vehicle's flight 3-DOF LCvx, with its own shortest-feasible t_f search.
- **C**: 3-DOF LCvx with SOC-representable thrust-rate and second-difference limits, sized to the vehicle's angular acceleration.
- **S**: standard 6-DOF successive convexification [17, 20] on the full quaternion rigid body. The whole nonlinear dynamics are linearized about the current trajectory (multiple shooting), with virtual control under an exact ℓ_1 penalty, an ∞-norm trust region and the usual actual-to-predicted ratio test.
- **B**: a planner built directly on Theorem 4.4, included to show that the split of (P2)–(P3) can be used as it stands, not as a recommended replacement for S. Its attitude variable is the body angular acceleration $\alpha(t)$ (pitch and yaw, piecewise linear, so the gimbal angle never jumps). With zero roll rate and axisymmetric inertia, the body rate is exactly linear in α, and the torque balance of Theorem 4.4 fixes the lateral thrust, $T_x = -(J_t/L)\alpha_y$, $T_y = (J_t/L)\alpha_x$. For a fixed α the problem in the axial thrust $T_a(t)$ is an exact SOCP. B alternates a convex trust-region step in (α, T_a), linearizing only the effect of the attitude change on the thrust direction, with an exact re-solve of this SOCP, and accepts a step only if the exact merit decreases. Every accepted iterate is therefore an exact trajectory of the planning model.
- **F**: the program (8) of Section 4.5, with $S = 16$ spans (63 spline coefficients) and the fixed references of Remark 4.9. One SOCP per plan: no initial trajectory, no linearization, no trust region, and nothing carried from one plan to the next. Its time of flight is the smallest t_f at which (8) is feasible, found by bisection (about ten SOCPs), plus 0.25 s. If a replan is infeasible or fails the slew-rate check of Remark 4.10, the previous plan is kept. By default F adds the terminal row $u_z(t_f) = g_0$ (arrive in hover) and drops it when that makes the program infeasible.
- **D** (reference): IPOPT on the full quaternion rigid body, multistart. **LB**: the 3-DOF relaxation at the same t_f, a lower bound on the fuel of any flyable plan, up to its discretization.

S, B and D share the model (inertia and lever arm frozen at their initial values), the function class (first-order-hold body thrust, $N = 35$), and the constraints: planned throttle 440–920 N inside the physical range, gimbal $|T_x|, |T_y| \le \tan 10^\circ \, T_z$, linear sufficient conditions for gimbal-rate and throttle-slew limits, tilt $\le 20^\circ$, 10° glide slope, upright at rest on the pad, and a gimbal that starts

at its current deflection. Every plan from S and B is judged by the same a posteriori certificate. It checks the physical limits on an 8× finer grid, then replays the planned body-frame thrust open loop through the full quaternion rigid body, which must land within 5 cm. Neither planner was tuned per scenario. F uses the same planning limits (with the 10° gimbal limit as a cone) and Assumption 4.6 in place of frozen inertia; its plans are checked on a dense grid and replayed in the same way.

Machine-checked ingredients. The elementary claims these planners rely on are formalized in Lean (28 theorems, namespace `LcvxSolver`; Table 2).

- *N1 (gimbal box)*: $|T_x|, |T_y| \le \tan\delta\, T_z$ gives $|\theta|, |\phi| \le \delta$ in the flight gimbal convention.
- *N2 (gimbal rate)*: with $B = \omega_g T_{\text{floor}} h$ and $t = \tan\delta_p$, the pair $|\Delta T_x| + t|\Delta T_z| \le B$, $|\Delta T_y| + t|\Delta T_z| \le B$ bounds both gimbal-angle rates by ω_g, including over first-order-hold intervals (`gimbal_rate_pair`, `foh_gimbal_rate`). Formalization caught an error: the originally stated stand-alone justification for the ϕ row is false (`phi_rate_heuristic_false`). The pair actually imposed is nevertheless valid.
- *N3–N5*: throttle slew, chord bound, and a direction-acceleration bound (continuous time only).
- *N6*: with zero roll torque and zero initial roll rate on an axisymmetric body, the gyroscopic term vanishes and the body rate is linear in the torque profile.
- *N7*: $V_3 \le V_5 \le V_6$ for any cost of the thrust history, also with free final time (`value3_le_value6_freeTime`), which makes LB a lower bound. Its grid discretization is not covered.

6.2 Results

- *(P1) False feasibility, visible in practice.* On a grid of 20 pad-hop targets (launch speeds 5–7 m/s, landing offsets 0–6 m, $t_f = 4$ s), the 3-DOF relaxation and the flight LCvx report "optimal" at all 20. None of B, S and IPOPT finds a flyable plan at 11 of them: all 8 targets with launch speed ≥ 6.5 m/s, plus 3 targets 6 m away. The three agree on feasibility in all 20 cells, and on fuel within 0.3% where they find a plan. "No plan found" is not a proof of infeasibility, but this is exactly the failure mode Theorem 3.1 predicts (Corollary 3.2). In-air, A's plans are not flyable either: 3 of 8 deliver 296–305 N on the "400 N" arc (lossiness at minimum throttle, as Theorem 4.2 predicts), and all demand instantaneous thrust re-pointing.
- *(P2) Iterating on the attitude reaches the 6-DOF optimum.* On 16 open-loop in-air problems (8 scenarios, each at B's minimum certified t_f and $t_f + 0.5$ s), every plan from B and S passed the certificate (replay landing within 1 mm). Both are within 0.03% of the best IPOPT optimum on every problem, and at most 0.85% above LB, so within 1% of the 6-DOF optimum at that t_f (Fig. 6). S converges from a straight-line initial guess as well as from the 3-DOF solution. The 3-DOF planner with rate limits does worse: C is infeasible at B's t_f in all 8 scenarios and uses 3.9–12.6% more fuel at its own minimum t_f.
- *(P3) The exact inner problem is useful but not decisive.* B, which exploits Theorem 4.4, ties with S on optimality and feasibility. It reaches a first flyable plan in a median 0.24 s of solver time, against 0.29 s for S from the same warm start. It solves many small SOCPs (36–108 variables) where S solves fewer, larger ones (about 650).
- *Closed loop.* The 6-DOF plant has wind, gusts, thrust noise, 5% Isp error and gimbal misalignment; all planners use a common tracker and 1 Hz replans, and B and S share the same replanning and fallback rules. Success means touchdown ≤ 1 m/s, lateral error ≤ 1 m and tilt $\le 5°$.

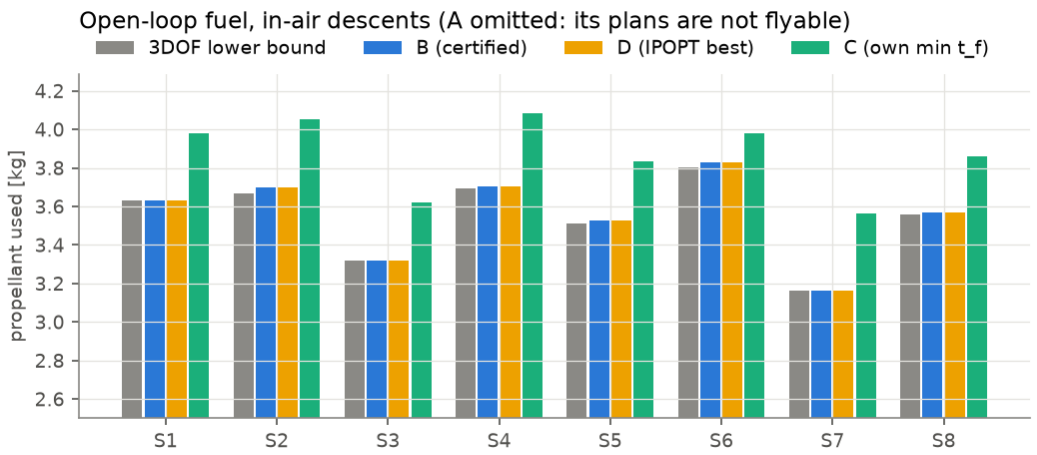


Figure 6: Open-loop fuel for 8 in-air descents: 3-DOF lower bound, planner B (certified), best IPOPT 6-DOF multistart, and 3-DOF with rate limits at its own minimum t_f. The standard SCvx planner S coincides with B and IPOPT to within 0.03% (not drawn). Baseline A is omitted because its plans are not flyable. Numerical.

	A (flight 3-DOF)	C (3-DOF + rates)	B	S (SCvx)
in-air success (40 runs)	28%	95%	100%	100%
median touchdown speed, in-air	2.42 m/s	0.48 m/s	0.20 m/s	0.13 m/s
pad-hop success (30 runs)	0%	100% (+23% fuel)	97%	100%
solver time per replan, mean (in-air)	0.007 s	0.004 s	0.33 s	0.62 s

Both 6-DOF planners land every in-air run, against 28% for the flight 3-DOF guidance (Fig. 7). S was as good as B or slightly better: it landed B's one failed hop, and its replans recovered from disturbances with fewer final-time extensions. B was about twice as fast per replan.

6.3 A sound single convex program in the loop

This subsection tests prediction (P4) . B and S were re-run alongside F on a second machine, so solver times here are not comparable with the table above; their success rates reproduce.

- *Open loop.* On the 8 in-air scenarios at B's minimum $t_f + 0.5$ s, one solve of (8) uses 0.10% more fuel than the better of B and S on average and 0.28% at most. It takes one SOCP and a median 0.046 s of solver time, against 9–37 SOCPs and 0.79 s for B and 25–35 SOCPs and 1.30 s for S. F's shortest feasible t_f is 0.04–0.22 s later than B's, with 0.5–2.5% more fuel there.
- *Exactness.* Fed only F's throttle and gimbal commands, with no feedback, the quaternion rigid body with depleting mass under Assumption 4.6 lands within 23 nm and 12 nm/s of the target (16 replays, half starting tilted 3° and rotating). This is Lemma E.1 seen numerically; the error is integration error.
- *Model error.* Each planner is exact in its own model. Replayed open loop against the simulator's mass table instead, plans of B and S miss by a median 0.43 and 0.45 m and plans of F by 0.03 m: the table keeps ε within 1% over a descent, while frozen inertia lets $J_t/(mL)$ drift by 6%.

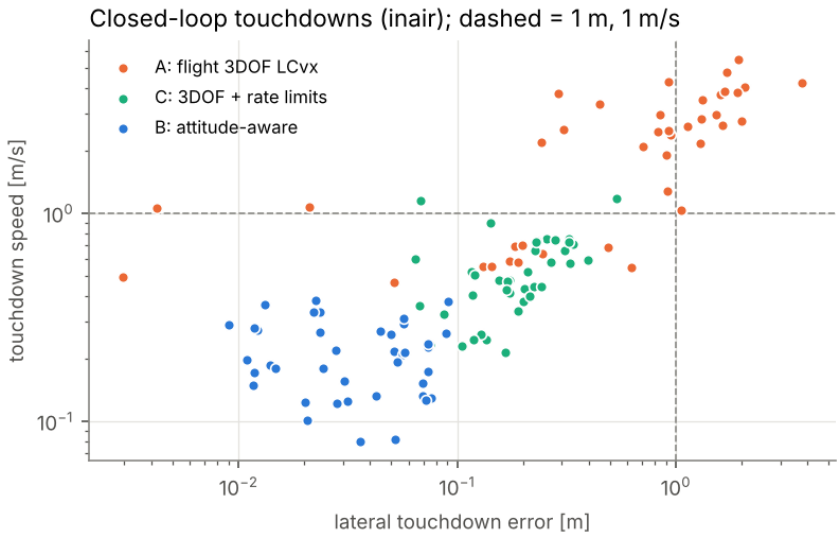


Figure 7: Closed-loop touchdowns, 40 in-air descents, planners A, B and C (dashed lines: 1 m, 1 m/s). S's touchdowns, not drawn, fall in the same region as B's (median 0.13 m/s and 3 cm, worst 0.37 m/s and 10 cm). Numerical.

- *Soundness and completeness on the hop grid.* On the 20 pad-hop targets of (P1), F is infeasible at all 11 targets where no flyable plan was found, and at 5 of the 9 where one exists; those 5 it reaches with 0.06–0.41 s more flight time. 3-DOF LCvx accepts all 20. In 600 solves of 300 random problems (default and random references), 458 were feasible and none violated a planning limit at any of 1201 instants.
- *The obstruction on this model.* With these planning limits, a 4 s hop launched vertically at 6.3 m/s can land 1, 2 or 3 m away, but IPOPT finds no plan that lands on the launch point, from symmetric starts or from nine asymmetric ones. The landable set has a hole, as Theorem 3.22 leads one to expect; this instance is numerical evidence, not a proof.
- *Closed loop, with computation time counted.* The earlier runs applied each plan at the instant its state was sampled. Here a plan computed from the state at time t becomes active at t plus its solver time, and each planner replans as soon as it can: B and S at most every 0.5 s, F every 0.1 s. Same plant, tracker, cases and seeds as above.

	B	S (SCvx)	F
In-air descents (40 runs)			
landed within limits	36	40	40
lateral error, median / worst	5.5 / 39.5 cm	5.9 / 18.3 cm	0.2 / 1.1 cm
touchdown speed, median / worst	0.23 / 1.65 m/s	0.23 / 0.58 m/s	0.19 / 0.29 m/s
solver time per replan, mean / worst	0.70 / 2.5 s	1.05 / 4.9 s	0.031 / 0.39 s
gimbal travel per flight, mean	141°	156°	45°
Pad hops (30 runs)			
landed within limits	28	30	30
lateral error, median / worst	2.9 / 11.1 cm	3.9 / 8.2 cm	0.4 / 2.0 cm
touchdown speed, median / worst	0.32 / 1.11 m/s	0.54 / 0.88 m/s	0.14 / 0.23 m/s
fuel at t_f, mean	1.69 kg	1.46 kg	1.54 kg

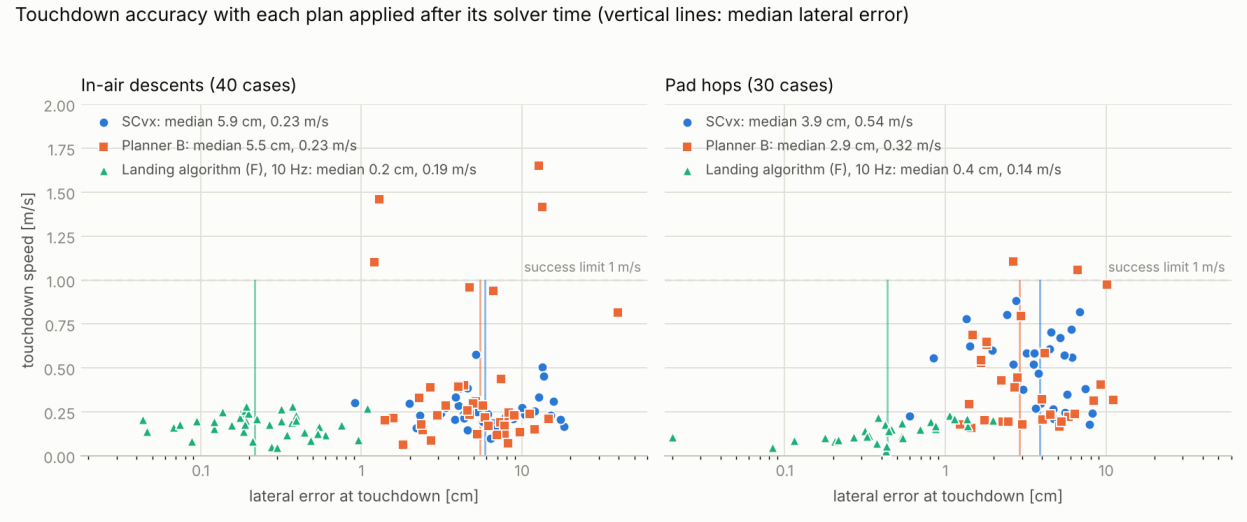


Figure 8: Closed-loop touchdowns with each plan applied after its solver time: standard SCvx (S), planner B and planner F, 40 in-air descents and 30 pad hops. Vertical lines mark median lateral error. Numerical.

With every solver time multiplied by four, S loses 4 of 30 hops and F, replanning every 0.2 s, loses none (Fig. 8).

- *Same replan rate, zero computation time.* Under the protocol of the previous subsection (1 Hz, plans applied instantly, all at B's t_f) lateral accuracy is a tie: median 3.1 cm for S and F in the air, 2.4 cm for both on hops. F touches down slower on hops (0.23 against 0.51 m/s) and never extends the final time in the air, where S does in 32 of 40 runs. The accuracy gain in the table therefore comes from replan rate, which F's solver time permits and the others' does not.
- *Ablations.* Giving S the same arrive-in-hover terminal condition does not close the gap (hops: touchdown 0.52 m/s, lateral 4.0 cm). Taking F's references from its previous plan (Remark 4.9) lowers in-air touchdown speed to 0.13 m/s and raises lateral error to 0.8 cm. With references and without the hover condition, F's in-air touchdown speed is 0.25 m/s.
- *Where F is worse.* Pad-hop fuel (+5% against S); minimum flight time; and, with zero computation time, in-air touchdown speed against S in 27 of 40 cases. Slew limits are outside Theorem 4.7: under 1% of replans failed the check and were not flown.

What this shows and does not show. The evidence is consistent with all four predictions. 3-DOF guidance produces plans the vehicle cannot fly and claims landings it cannot reach. Planners that iterate on the attitude reach the 6-DOF optimum and land reliably. The exactness of Theorem 4.4 can be built into such a planner directly. It does not, on these tests, make B better than standard successive convexification. Iteration is not necessary for flyable plans: the single program (8) gives up 0.1% of fuel at the operating point and a few tenths of a second of minimum flight time, never returned an unflyable plan, and, because it is fast enough to replan ten times a second, landed more accurately than either iterative planner once computation time was counted. Not shown: convergence of B or S, global optimality (only the gap to LB is bounded), timing on flight hardware, actuator lag and state-estimation error, a vehicle whose centre of percussion moves by more than the 1% of the simulator's mass table, and robustness beyond the Monte Carlo cases.

7 Machine-checked formalization

The obstruction is short but easy to misstate: data entry, terminal attitude, state constraints and the sign conventions of the dive-then-brake face all matter. We therefore formalized its core in Lean 4 with Mathlib [29, 30] (toolchain `v4.36.0-rc2`; project `lean/LcvxObstruction` in the supplementary archive, namespace `LcvxObstruction`). The project contains no `sorry` and no `axiom`. For every Lean declaration cited in this paper (126 names, including every headline theorem), `#print axioms` reports exactly

`[propext, Classical.choice, Quot.sound]`,

the standard axioms of classical Lean mathematics. The script `Verify.lean` in the project root repeats this check: after a full `lake build`, `lake env lean Verify.lean` prints the headline statements and fails if any cited declaration is missing or depends on any other axiom (a `sorry` would appear as the axiom `sorryAx`).

Headline theorem. The Main Theorem of Section 1 is `no_lossless_relaxation_constrained_5dof` together with its LCvx (optimization) form `no_losslessFor_relaxation_constrained_5dof`, the upright-landing version `no_lossless_relaxation_constrained_upright_5dof`, and the kinematic 6-DOF versions `no_lossless_relaxation_constrained_6dof` and `no_lossless_relaxation_constrained_upri` Their hypotheses are: constant acceleration bounds $0 < a_{\min} \le a_{\max}$ with $g_0 < a_{\max}$, $0 < \delta < \pi/2$, $\omega > 0$, any $t_f > 0$, $\tan\gamma \ge 0$, and any state-constraint set containing the glide-slope cone $\{r_z \ge 0,\ \tan\gamma\|r_h\| \le r_z\}$. The 6-DOF attitude is a path of proper rotations $A(s)$ with $A(0)e_z = e_z$ in which every body-fixed direction turns at rate at most ω (total body rate $\le \omega$), with free roll angle. The non-vacuity lemma `constrainedLandingSet_nonempty` shows the constrained set is nonempty. The same statements hold with explicit mass depletion (`..._mass` versions, with non-vacuity `constrainedLandingSet_mass_nonempty`), on free-time windows, and at the vehicle's constants. The constants are checked by `norm_num`: the burn $\alpha\rho_2 \cdot 42 \approx 23.8 \le 24$ kg, and $g_0 < \rho_2/m_0$.

7.1 What is formalized

Table 2 lists the formal theorems that correspond to statements in this paper. A row-by-row faithfulness map from each Lean definition to its mathematical meaning is maintained in `lean/MODEL.md`. Its main conventions are listed next.

- $\mathbb{R}^3$ is `EuclideanSpace ℝ (Fin 3)`. Angles are Mathlib's `InnerProductGeometry.angle`. Integrals are Bochner integrals over $[0, t_f]$.
- `UniformConvexRelaxation Θ Y` is a convex set $C \subseteq \Theta \times Y$ for arbitrary real vector spaces Θ, Y. `IsLossless F` is the feasibility form of Definition 2.4(a), and `IsLosslessFor J F` is the optimization (LCvx) form (b) for an arbitrary cost J, including attainment.
- `AttitudeAdmissible tf ω δ d b` encodes M1 with variant (F): $b(0) = e_z$, unit vectors, $\angle(b(s), b(s')) \le \omega|s - s'|$, measurable unit d, gimbal $\angle(d, b) \le \delta$. Pointwise constraints are imposed for every s, which changes no integral. The admissible thrust accelerations form an abstract class `Acc`. Theorem B assumes only $a \ge a_1$ for a continuous $a_1 > 0$ (so it covers mass depletion through $a_1(s) = \rho_1/(m_0 - \alpha\rho_1 s)$). Theorem B* assumes a box $a_{\min} \le a \le a_{\max}$ with constant bounds, which is exact for constant mass.

- Positions and velocities are given in closed (Cauchy) form, $r(t) = r_0 + tv_0 + \frac{t^2}{2}g + \int_0^t (t-s)\,a\,d$, and `position_eq_integral_velocity` proves that this equals $r_0 + \int_0^t v$.
- `AccMass` is the explicit mass-depletion model ($\dot{m} = -\alpha T$, $a = T/m$, dry-mass limit); `accMass_spec` derives $a \geq a_1$ from it.
- `StateLinearRelaxation` is Definition 2.6 with an affine readout and a convex penalty that is nonnegative on C.

7.2 The PMP-as-hypothesis convention

Two external theorems appear on the positive side, and each enters only as an explicit hypothesis of the formal statement, never as an axiom. The Pontryagin maximum principle is a structure `PMPConclusion` whose fields are the adjoint equations and the a.e. minimum condition of (H-PMP). Aumann's theorem enters as `AumannInstance`, the single instance used in Theorem 4.1. A reader can therefore check exactly which conclusions depend on which external results. The positive-side files (`Positive/`: pointwise cap lemmas, primer geometry, Theorem 4.2 under `PMPConclusion`, discrete KKT and purification, Theorem 4.4 pointwise and trajectory-level convexity, Theorem 4.1 feasibility under `AumannInstance`) are summarized in `lean/POSITIVE.md`. There the Lean form is weaker than the paper in several places, which that file lists explicitly: the adjoint ODEs are assumed globally differentiable; C-S(2) is constant-mass only; "P_0-optimal" is reduced to pointwise feasibility; C-E is discrete-time; and C-W is constant-mass and unconstrained.

7.3 What is not formalized

Theorem 4.7 and Proposition 4.8: only the first step of the proof, the centre-of-percussion identity (`TVC.cop_identity`), is in Lean, and there with constant mass; the flat map, the boundary equivalences, the pointwise bounds, the control-polygon argument and the mass bracket (Lemmas E.1 to E.4) are proved on paper and tested numerically. They are elementary and would be a natural next formalization target. Also not formalized: the tent variant (T) of the unconstrained Theorem B; the envelope version B*-env; unbounded free final time, and the instantiation of the line-search lemma from B*; the ball bound Theorem 3.5 and ε-losslessness; torque-limited 6-DOF models; the pure-TVC certificates themselves (the analytic reduction, the witnesses and the exact-arithmetic numerics enter `tvc1_main` as hypotheses) and a trajectory-level pure-TVC landing set with varying $\varepsilon(m)$; $\rho_1 = 0$; the SDP results; Cor. D.1; Lemma S beyond the single explicit fixed-thrust instance (throttle, gimbal $\delta > 0$, mass depletion, glide slope), and the model facts behind Proposition T. The physical scope caveat of Section 3.3 also applies to the formal Theorem B: its witnesses start below ground for a vehicle that cannot hover at minimum throttle. The formal Theorem B* is the statement that covers the altitude-constrained problem.

8 Discussion

8.1 Implications for guidance design

1. *Do not expect a single LCvx-type SOCP for attitude-rate-limited landing.* Within the class of convex programs on the physical state with affine data entry, exactness is impossible, and this cannot be fixed by adding slacks, cuts, lifted variables, discretization or a line search over t_f (Theorems 3.1, 3.7 and 3.12 and Lemma 3.10); the state-constrained core of this statement is machine-checked. Any such program, if it contains all genuine trajectories, accepts boundary

data that no flyable trajectory can land from. The flight software then has no way to detect infeasibility from the convex program alone. The statement is about exactness. It does not say that a single SOCP cannot be *sound*; see item 3.

2. *Outer attitude loop, exact convex inner problem.* The nonconvexity lies entirely in the choice of attitude profile (Theorem 4.3). For a fixed attitude profile, the time-varying-cone LCvx is exact (Theorems 4.1 and 4.2). On a pure-TVC vehicle the problem given the attitude is *exactly* convex, with no relaxation (Theorem 4.4). A natural architecture is therefore an outer search over a low-dimensional family of attitude profiles (or a sequential update of b) around an exact convex inner problem. Every inner solution is then genuinely flyable, and the inner problem certifies infeasibility for the given profile. Section 6 implements this architecture and evaluates it numerically. Standard successive convexification, which linearizes the attitude together with everything else, performed as well, so the architecture's value is speed and flyable intermediate iterates rather than better plans.

3. *Or no loop at all: a sound single SOCP.* On a pure-TVC vehicle the outer loop exists only because the problem is posed in centre-of-mass coordinates, where attitude and thrust multiply. In centre-of-percussion coordinates the dynamics are linear, and restricting the actuator limits instead of relaxing them gives one SOCP whose plans are all flyable at every instant (Theorem 4.7). What it gives up is measured in Section 6.3: 0.1% of fuel at the operating point and a few tenths of a second of minimum flight time, concentrated in manoeuvres that tilt in order to shed thrust. Because it needs no initial trajectory, trust region or iteration budget and solves in tens of milliseconds, it can replan at the control system's own time scale, and in our simulations that mattered more for touchdown accuracy than the last fraction of a percent of optimality. For a new gimballed vehicle we would start here and keep successive convexification for the manoeuvres this program refuses.

4. *Count computation time.* Closed-loop comparisons that apply each plan at the instant its state was sampled favour slow planners. With plans applied after their solver time, the planner built on Theorem 4.4 lost 4 of 40 in-air landings it had made before, and the lateral error of successive convexification roughly doubled.

5. *Certificates.* The degree-2 SDP lift gives a rigorous lower bound on the minimum landing error and certifies infeasibility of exact landing when positive. Its global optimality can be certified a posteriori (rank 3, stress test), and a priori for coplanar targets (Theorems 4.12 and 4.13). It is a useful companion to a local or sequential solver, not a replacement.

6. *How much does it matter?* For in-air descents of a small lander, the exhibited nonconvexity is centimetre-scale, and a sound convex program loses little in value (0.1% of fuel in Section 6.3). The large effects are elsewhere. First, near lift-off (pad-launched hops) the landing set can have a metre-scale hole. Second, 3-DOF LCvx solutions ask for thrust-direction changes far beyond the vehicle's angular acceleration (Section 5). Third, on a pure-TVC vehicle the 3-DOF optimum is essentially never exactly flyable. The "upright body plus pointing cone" 3-DOF model has effective cone zero.

7. *Tilt-angle rate limits.* Constraining only the tilt angle from vertical, with free azimuth, removes rigidity and appears to give convex landing sets numerically. It is not SOC-representable either, and it is not the physical constraint of a rigid body.

8.2 Open problems

1. *Pure TVC, upright landing at rest, with altitude constraint.* This is the actual small-lander case. It is proved, computer-assisted and certified in exact arithmetic, with constant mass and unlimited gimbal rate, at $t_f = 1,2$ s without a tilt cap and at $t_f = 3,5$ s in the tilt-capped class (Theorem 3.22). The fully realistic model (rate-limited gimbal, mass depletion, free roll) is nonconvex numerically at every horizon tried ($t_f = 3$–7 s, gaps 0.5–3.3 cm; a 1.7 m hole at hop data). Still open: certificates with a gimbal-rate limit, with mass depletion, at $t_f = 7$ s capped (fails by 4.8%), for uncapped $t_f > 2.3$ s (tumbling lemma), for the box gimbal and free roll beyond $t_f = 1$ s, and with free final time.

2. *Rigor at realistic constants.* For our vehicle, Lemma S (hard soft landing, per instance) and the hop-instance hole are numerical. The available quadratic, Wirtinger and S-procedure certificates are provably too loose. A two-dimensional rigidity argument or validated branch-and-bound is needed. The $\rho_1 = 0$ landing-set result with altitude constraint is open at $T/W \approx 1.27$.

3. *CSD2′.* Is the degree-2 lift exact in continuous time with fixed thrust, or for the convexified-lens and $\delta = 0$ discretizations? Is there a tractable exact lift with variable throttle? (The natural degree-2 throttle lift and the separate-axis lift CSD3 were both found lossy numerically.)

4. *Minimum-fuel value-losslessness.* Does V agree with its convex envelope $\operatorname{conv} V$ on F for minimum fuel? Theorem A-opt reduces this to a convexity question, and numerics show no violation. A positive answer would still not give an LCvx in the detection-sound sense.

5. *Long horizons and coning.* At long horizons with tilt caps, coning fills most but not all of the dents. Whether nonconvexity persists for all horizons in the tent variant, and at faces with long capped plateaus, is open.

6. *Quantitative state-constrained gaps.* The rigorous B* gap ($\sim 10^{-5}$ m) is far below the numerical one ($\sim 10^{-2}$). A sharp quantitative version would settle how much a guidance designer actually loses.

7. *The sound program of Theorem 4.7.* (a) Formalize it; every lemma is elementary. (b) Bring the gimbal and throttle slew limits inside the guarantee: bounds with worst-case constants are valid but too loose to be useful, and a restriction with reference parameters, as in rows (R3)–(R5), is the natural route. (c) Bound the loss: Proposition 4.8 says the program is incomplete, and the experiments say by how much on one vehicle, but there is no theorem bounding the fuel or flight-time penalty. (d) Remove Assumption 4.6: with ε varying along the flight the output y is no longer flat, and an extra term $2\dot{\varepsilon}\dot{b}$ appears in $\ddot{y}$. (e) Is there a larger convex sound set, for instance one that allows tilting to shed thrust in a fixed direction?

Reproducibility. The supplementary archive contains the Lean project (`lean/`: sources, `Verify.lean`, and the notes `MODEL.md`, `POSITIVE.md`, `SLICE.md`, `SOLVER.md` and `TVC.md`, which map each Lean definition to its mathematical meaning) and the scripts, data and derivation notes of the computer-assisted pure-TVC certificates (`certificates/`), and the code, tests and proof note of the sound single convex program of Section 4.5 (`landing/`). Its `README` gives the commands that reproduce every check.

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A Proofs of Theorems B and Q

Throughout, M1 of Section 2.2.

Lemma A.1 (Spherical triangle inequality, inner-product form)**.** *Let $x, y, z \in S^2$, $\alpha, \beta \in [0, \pi]$, $\alpha+\beta \le \pi$, $\langle x, y\rangle \ge \cos\alpha$, $\langle y, z\rangle \ge \cos\beta$. Then (a) $\langle x, z\rangle \ge \cos(\alpha+\beta)$. (b) If $\langle x, z\rangle = \cos(\alpha+\beta)$ and $\alpha+\beta < \pi$, then $\langle x, y\rangle = \cos\alpha$, $\langle y, z\rangle = \cos\beta$, and, if $\beta \in (0, \pi)$, $x = \cos\alpha\, y - \frac{\sin\alpha}{\sin\beta}(z - \cos\beta\, y)$.* [Lean-checked]

Proof. Write $p = \langle x, y\rangle = \cos a$, $q = \langle y, z\rangle = \cos c$ with $a \le \alpha$, $c \le \beta$, and $x' = x - py$, $z' = z - qy$, both orthogonal to y. Then $\langle x, z\rangle = pq + \langle x', z'\rangle \ge \cos a\cos c - \sin a \sin c = \cos(a + c) \ge \cos(\alpha + \beta)$. Equality forces $a = \alpha$, $c = \beta$ (strict monotonicity of cos on $[0, \pi]$) and $\langle x', z'\rangle = -\|x'\|\|z'\|$, i.e. x' is a nonpositive multiple of z'. □

Corollary A.2 (Extremal thrust direction)**.** *If $\langle d, b\rangle \ge \cos\delta$, $\langle b, e_z\rangle \ge \cos\varphi$, $\varphi \in (0, \pi)$, $\delta + \varphi < \pi$, then $d_z \ge \cos(\delta + \varphi)$. Equality implies $b = \cos\varphi\, e_z + \sin\varphi\, u$ for a horizontal unit u and $d = \cos(\delta + \varphi)e_z + \sin(\delta + \varphi)u$.*

Lemma A.3 (Rigidity)**.** *If $b(s) = \cos\psi\, e_z + \sin\psi\, u$, $b(s') = \cos\psi'\, e_z + \sin\psi'\, u'$ with $\psi, \psi' \in (0, \pi)$, u, u' horizontal unit vectors, and $\langle b(s), b(s')\rangle \ge \cos(\psi - \psi')$, then $u = u'$.* [Lean-checked]

Proof. $\langle b(s), b(s')\rangle - \cos(\psi - \psi') = \sin\psi \sin\psi'(\langle u, u'\rangle - 1) \ge 0$ forces $\langle u, u'\rangle = 1$. □

Proof of Theorem 3.4. The extreme arcs (4) are admissible: $\langle b_u(s), b_u(s')\rangle = \cos(\varphi(s) - \varphi(s'))$ and φ is ω-Lipschitz; the gimbal angle is δ; $a = a_1$ at constant minimum throttle. So $P_{\pm u} \in R$ and $M \in \operatorname{conv} R$. Suppose $M = \int a\, d$ for an admissible (T, d, b). In both variants $\omega t_f \le \pi$ and $0 \le \delta + \varphi \le \pi/2$.

Step 1. (AL_ω) with $s' = 0$ (and $s' = t_f$ in variant T) gives $b_z(s) \ge \cos\varphi(s)$. By Lemma A.1(a), $d_z \ge \cos(\delta + \varphi) \ge 0$, and with $a \ge a_1$, $a\, d_z \ge a_1 d_z \ge a_1 \cos(\delta + \varphi)$ a.e. *Step 2.* Integrating, $M_z \ge \int a_1 \cos(\delta + \varphi) = M_z$, so the nonnegative integrand $a\, d_z - a_1 \cos(\delta + \varphi)$ vanishes a.e. Since $a_1 > 0$, $d_z = \cos(\delta + \varphi)$ a.e. *Step 3.* By Corollary A.2, for a.e. $s \in (0, t_f)$, $b_z(s) = \cos\varphi(s)$ and $d(s) = \cos(\delta + \varphi)e_z + \sin(\delta + \varphi)u(s)$, where $u(s)$ is the azimuth of $b(s)$. *Step 4.* $b_z - \cos\varphi$ is continuous and vanishes a.e., hence everywhere, so $u(\cdot)$ is defined and continuous on $(0, t_f)$. *Step 5.* In variant F, for $0 < s < s'$, (AL_ω) gives $\langle b(s), b(s')\rangle \ge \cos(\varphi(s') - \varphi(s))$, and Lemma A.3 gives $u(s) = u(s')$. In variant T apply this on $(0, t_f/2]$ and on $[t_f/2, t_f)$ and chain through $t_f/2$. So $u \equiv u_0$. *Step 6.* $I_h = (\int a \sin(\delta + \varphi))u_0$ with coefficient $\ge \int a_1 \sin(\delta + \varphi) > 0$, contradicting $M_h = 0$. Nonconvexity of F_{t_f} follows from Lemma 3.3, and impossibility from Theorem 3.1. □

Lemma A.4 (Support bound)**.** *For any admissible control (no boundary condition on b needed) and τ with $\delta + \omega \max(\tau, t_f - \tau) \le \pi/2$, $\|I\| \ge \langle I, b(\tau)\rangle \ge G(\tau) = \int_0^{t_f} a_1(s)\cos(\delta + \omega|s - \tau|)\, ds$.*

Proof. (AL_ω) gives $\langle b(s), b(\tau)\rangle \ge \cos(\omega|s - \tau|)$, and Lemma A.1(a) gives $\langle d(s), b(\tau)\rangle \ge \cos(\delta + \omega|s - \tau|) \ge 0$. Multiply by $a \ge a_1$ and integrate. □

Proof of Theorem 3.5. For $I \in R$, $\|I - M\| \ge \|I\| - \|M\| \ge G(t_f/2) - M_z$ by Lemma A.4. With $c(x) = \cos(\delta + \omega x)$, strictly decreasing on $[0, t_f]$, put $h(s) = c(|s - t_f/2|) - c(s)$. Then $h \ge 0$ for $s \ge t_f/4$ and $h < 0$ for $s < t_f/4$. Since a_1 is nondecreasing, $(a_1(s) - a_1(t_f/4))h(s) \ge 0$, so $G(t_f/2) - M_z = \int a_1 h \ge a_1(t_f/4) \int h$. A direct computation, using $\sin x + \sin(x + 2k) = 2\sin(x + k)\cos k$, gives $\int_0^{t_f} h = \frac{4}{\omega}\sin(\delta + \frac{\omega t_f}{2})\sin^2\frac{\omega t_f}{4}$. The sign condition holds if $\delta + \omega t_f \le \pi/2$ (nonnegative integrand), or if $\alpha = 0$, where $M_z = \frac{a_1}{\omega}(\sin(\delta + \omega t_f) - \sin\delta) \ge 0$. The relaxation statement is Corollary 3.2(ii) together with Lemma 3.3. □

B Proof of Theorem B*

Notation: $a_{10} = \rho_1/m_0$, $a_{\max} = \rho_2/(m_0 - \alpha\rho_2 t_f)$, $A_b = \rho_2/m_0 - g_0$, $w(s) = s - s^*$. For a control c put $W(c) = \int w\, a\, d$, $X(c) = \int_0^{s^*}(s^* - s)a\, d$, $B(c) = \int_{s^*}^{t_f}(s - s^*)a\, d$, so $W = -X + B$. By (5), $\Lambda(\theta) = W(c) + \kappa e_z$ with $\kappa = g_0 t_f(s^* - t_f/2)$, for any control c landing from θ.

Proof of Theorem 3.7 (qualitative part). Step 1 (direction bound for a fixed throttle). Fix T, hence $a = a_T > 0$. On $[0, s^*]$, $b_z \geq \cos\omega s$ and $\delta + \omega s \leq \pi/2$ by (S1), so $d_z \geq \cos(\delta + \omega s) \geq 0$. On $(s^*, t_f]$, $d_z \leq 1$. Since $w < 0$ before s^* and $w > 0$ after,

$$W_z(c) \leq V(T) := -\int_0^{s^*}(s^* - s)a_T \cos(\delta + \omega s)\, ds + \int_{s^*}^{t_f}(s - s^*)a_T\, ds.$$

Step 2 (attained for every T). The dive-then-brake control c_u^T uses $b = \cos(\omega s)e_z + \sin(\omega s)u$ and $d = \cos(\delta + \omega s)e_z + \sin(\delta + \omega s)u$ on $[0, s^*]$, then $b \equiv b(s^*)$ and $d = e_z$. It is admissible, because by (S1) $\angle(e_z, b(s^*)) = \omega s^* \leq \delta$. In variant (T′), return b meridionally to e_z on $[s^*, 2s^*]$. The gimbal angle stays $\leq \delta$. *Step 3 (late throttle).* Replacing T by ρ_2 on $(s^*, t_f]$ leaves a unchanged before s^* and does not decrease it after. So $V(T) \leq V(T^\sharp)$, strictly unless $T = \rho_2$ a.e. late. *Step 4 (outer maximum).* $V^* = \sup_T V(T)$ is attained, explicitly for $\alpha = 0$ ($T^* = \rho_1$ then ρ_2), and for $\alpha > 0$ by weak-$*$ compactness of the throttle set and weak-$*$ continuity of V. *Step 5 (state constraints).* On the brake $d = e_z$, so $v_h = r_h = 0$ there and $\dot v_z \geq A_b$. Hence $v_z(t) \leq -A_b(t_f - t)$ and $r_z(t) \geq \frac{1}{2}A_b(t_f - t)^2$. On the dive $\dot v_z \geq -g_0$, so $v_z < 0$ by (S2). Thus r_z is decreasing, hence positive before t_f. Laterally $\|r_h\| \leq \frac{1}{2}a_{\max}s^{*2}$, and (S3) gives the glide slope. By rotation invariance, $R_\pi\theta^*$ is landed by $c_{-u}^{T^*}$. *Step 6 (midpoint infeasible).* $\Lambda(\theta_m) = (V^* + \kappa)e_z$. If $\theta_m \in F_{t_f}$ via c, then $V^* = W_z(c) \leq V(T) \leq V^*$, so all inequalities are equalities a.e. On the dive, $d_z = \cos(\delta + \omega s)$, so Corollary A.2 and Lemma A.3 give a constant azimuth u_0. On the brake, $d = e_z$. Then $W_h(c) = -(\int_0^{s^*}(s^* - s)a_T \sin(\delta + \omega s))u_0 \neq 0$, contradicting $W_h(c) = \Lambda_h(\theta_m) = 0$. □

Proof of the quantitative bound ($\delta \leq \pi/4$). Let c land from $\theta \in F_{t_f}$ and put $\eta = V^* - W_z(c) \geq \eta_{dive} + \eta_{brake}$, the deficits relative to Step 1. Let $\tau = \vartheta s^*$ with $\vartheta = a_{10}\sin\delta/(4a_{\max}) \leq \frac{1}{4}$, and $c_\tau = b(\tau)$, of tilt $\beta \leq \omega\tau$. (i) As in Lemma A.4, $\langle X, c_\tau\rangle \geq G(T) := \int_0^{s^*}(s^* - s)a_T \cos(\delta + \omega|s - \tau|)$, since $\delta + \omega s^* \leq \pi/2$. (ii) With $c_\tau = \cos\beta\, e_z + \sin\beta\, n$: $\sin\beta\|X_h\| \geq G - X_z \geq G - Z_0 - \eta_{dive}$, where $Z_0 = \int_0^{s^*}(s^* - s)a_T\cos(\delta + \omega s)$. (iii) The mean value theorem gives $G - Z_0 \geq \omega\tau\gamma_X$, using $(1 - \vartheta)^2 - \frac{1}{4} \geq \frac{5}{16}$. (iv) Since $\sin\beta \leq \omega\tau$: $\|X_h\| \geq \gamma_X - c_1\eta_{dive}$ with $c_1 = 1/(\omega\tau)$. (v) $\|d_h\| \leq \sqrt{2(1 - d_z)}$ and Cauchy–Schwarz give $\|B_h\| \leq \sqrt{2W_b\eta}$. (vi) $\|\Lambda(\theta) - \Lambda(\theta_m)\|^2 = \|W_h\|^2 + \eta^2$ and $\|W_h\| \geq \gamma_X - c_1\eta - \sqrt{2W_b\eta}$, so $\|\Lambda(\theta - \theta_m)\| \geq \eta_0$. Finally, $\|\Lambda\| = \sqrt{1 + s^{*2}}$. □

Proof of the free-final-time window variant. Take $W = [t_a, t_b]$ with $s^* < t_a$, (S1)–(S3) at t_a with $a_{\max}$ evaluated at t_b (they are monotone in t_f), and, under a dry-mass constraint, $m_0 - \alpha\rho_2 t_b \geq m_{dry}$ so that the full-throttle brake of every witness is fuel-feasible. $\ell^*(t) = \sup_{F_t}\ell = V_t^* + \kappa(t)$ is continuous in t, so it attains its maximum on $[t_a, t_b]$ at some t°. If the union were convex, $\theta_m(t^\circ) \in F_{t'}$ for some t' with $\ell(\theta_m) \geq \ell^*(t')$. Then θ_m would be a maximizer at horizon t', and Step 6 at that horizon forces $\Lambda_h \neq 0$. Note that Λ does not depend on the horizon. □

C Proof of Theorem S (Lemma S in a regime)

Model M1, $b(t_f)$ free, $T := t_f$, $\Omega = \omega T$, $\tau(s) = \delta + \omega s$, $\bar\tau = \delta + \Omega$, $a_{10} = \rho_1/m_0$, $a_2 = \rho_2/(m_0 - \alpha\rho_2 T)$, $\varepsilon_m = a_1(T)/a_{10} - 1$. The lateral force is $F = a\, d_h$, and the *gain* of a control is $\int a_1 - I_z$, the vertical impulse wasted relative to upright minimum throttle. P is the $L^2((0, T); \mathbb{R}^2)$ projection onto affine

functions. It depends only on the moments $m(F) = (\int F, \int sF)$, and π_m denotes the affine function with moments m. In particular $\|\pi_{(0,q)}\|^2 = 12|q|^2/T^3$.

Landing from vertical data. From $\zeta = (he_z, v_{0z}e_z)$ a trajectory lands at $(p, 0)$ with $v(T) = 0$ iff $I_h = 0$, $I_z = g_0T - v_{0z}$, $\int sF = -p$ and J_z is fixed by ζ. Hence *every trajectory from ζ has the same gain*, and $m(F) = (0, -p)$.

Proof of Lemma 3.13. Let $\tilde{f} = f - f(0) - \frac{s}{T}(f(T) - f(0))$. Then $f - Pf = \tilde{f} - P\tilde{f}$, and $\|\tilde{f} - P\tilde{f}\| \le \|\tilde{f} - \bar{\tilde{f}}\|$ for the mean $\bar{\tilde{f}}$. Since $\tilde{f}(0) = \tilde{f}(T) = 0$, $\tilde{f}$ extends periodically, and the periodic Wirtinger inequality (Parseval) gives $\|\tilde{f} - \bar{\tilde{f}}\|^2 \le (T/2\pi)^2\|\dot{\tilde{f}}\|^2 = (T/2\pi)^2(\|\dot{f}\|^2 - |f(T) - f(0)|^2/T)$. All steps are componentwise. □

Master inequality. With $c_0 = 1/(1 + \cos\delta)$, $\delta_e = 2\sin(\delta/2) + \varepsilon_m \sin\bar{\tau}$, $E_4 = a_{10}(1 + \varepsilon_m)^2 \int_0^T (\frac{1}{1+\cos\tau(s)} - c_0)\sin^2\tau(s)\,ds$, $B = a_{10}c_0T(\frac{\Omega}{2\pi} + \delta_e)^2 + E_4$, $A' = c_0/a_{10}$, $\kappa_0 = \cos\bar{\tau}/((a_2 - a_{10})\sin^2\bar{\tau})$ and $C_m = A'\kappa_0/(\kappa_0 - A')$ ($C_m = A'$ if $\rho_1 = \rho_2$):

Proposition C.1 (Master inequality)**.** *If $\bar{\tau} < \pi/2$ and $\kappa_0 > A'$, every admissible M1 trajectory (any 3-D attitude path, throttle, $\alpha \ge 0$) satisfies* gain $\le B + C_m\|\pi_{m(F)}\|^2$. *[proved; not machine-checked] [Lean-checked: constant mass, variant with constant $\bar{c}(T)$]*

Proof sketch. (1) A pointwise majorant $-a\,d_z \le -a_1 + a_1\bar{c}|d_h|^2 - (a - a_1)\cos\tau$ with $\bar{c} = 1/(1+\cos\tau)$ splits off the throttle surplus $K = (a - a_1)d_h$, which costs at least $\kappa_0\|K\|^2$. (2) Normalize the mass. (3) Eliminate K through its projection ($PK = \pi_m - a_{10}Pv$) by maximizing a concave quadratic. (4) The only nonconvex step bounds the non-affine part of the lateral tilt $\beta = b_h$, which is ω-Lipschitz, by Lemma 3.13: $\|P^\perp\beta\| \le \omega T^{3/2}/(2\pi)$. The gimbal and mass corrections are bounded pointwise by δ_e. □

Witness. A planar single-crossing S-turn at minimum throttle: tilt $\varphi(s) = \omega s$ for $s \le t_1$ and $\omega(2t_1 - s)$ after, the thrust deflected outward by δ ($\chi = \varphi + \delta\,\mathrm{sgn}\,\varphi$), and t_1 chosen by the intermediate value theorem so that $\int a_1 \sin\chi = 0$ (so $I_h = 0$). Its mean-square lateral tilt is $(3\sqrt{2} - 4)/6 \cdot \Omega^2 \approx 0.0404\,\Omega^2$ in the small-angle limit ($t_1 = (1 - 1/\sqrt{2})T$). This exceeds the Wirtinger cap $\Omega^2/(4\pi^2) \approx 0.0253\,\Omega^2$. The witness does not return to zero tilt, which is why it beats every path whose two lateral moments vanish.

Proof of Theorem 3.14. (1) Along the witness $\dot{v}_z = a_1\cos\chi - g_0 > 0$ by (S1), so $v_z < 0$ on $[0, T)$ and $r_z > 0$ before T. Rotations about e_z preserve ζ_w, so $L(\zeta_w)$ contains the circle $|p| = |p_w|$. (2) Every trajectory from ζ_w landing at $(p, 0)$ has gain gain$_w$ and $m(F) = (0, -p)$, so Proposition C.1 gives gain$_w \le B + 12C_m|p|^2/T^3$, i.e. $|p| \ge r_S$. (3) Apply Theorem 3.12 with $\varphi(x) = \|x_r\| + \iota_{\{x_v = 0\}}(x)$; since $0 \in \mathrm{conv}\,L(\zeta_w)$, the relaxed value is 0. □

For $\delta > 0$ the certified range of Ω has a positive lower end: as $\Omega \to 0$ at fixed δ, gain$_w \to a_{10}T(1 - \cos\delta)$ while $B \to a_{10}T \cdot 2\sin^2(\delta/2) \cdot 2/(1 + \cos\delta)$, which is larger, so $M < 0$. At $T = 2$ s the certified ranges are $(0, 0.55]$ ($\delta = 0$), $[0.004, 0.46]$ (5°), $[0.009, 0.42]$ (7°), $[0.038, 0.34]$ (10°) and $[0.070, 0.27]$ (12°) (independent re- computation). All rows of Table 3 lie inside them. With $\alpha = 5.66 \times 10^{-4}$ the $\delta = 5°$ row fails narrowly, because ε_m enters δ_e linearly. That is a weakness of the constants, not of the geometry. For $\delta \ge 15°$ the analytic constants do not certify; a discretized S-procedure SDP suggests $\delta = 15°$ holds with an 8% margin (numerical).

D Further proof sketches

D.1 Theorem P0-F ($\rho_1 = 0$)

Set $\rho_1 = 0$, $\alpha = 0$, $a_2 = \rho_2/m_0$, units $\omega = a_2 = 1$, $T = \omega t_f$. For fixed b the pointwise control set $U(b) = \{u : \|u\| \le a_2,\ \langle u, b\rangle \ge \|u\| \cos\delta\}$ (convex, containing 0) has support function $a_2\|p\|\, c(\angle(p, b))$ with $c(x) = 1$ for $x \le \delta$, $\cos(x-\delta)$ for $\delta \le x \le \delta + \pi/2$, and 0 beyond. K is compact and star-shaped about 0, so it is convex iff every exposed face lies in K.

Take $\hat{p} = (\cos\delta, 0, -\sin\delta)$ and $\sigma \in (0, t_f)$, primer $p(s) = (s-\sigma)\hat{p}$, face functional $\ell_\sigma = \langle \hat{p}, \int (s-\sigma) a\, d\rangle = \langle \hat{p}, r_0 + \sigma v_0\rangle + \text{const}$. With $\beta = \phi_0 - \angle(\hat{p}, b)$ (an ω-Lipschitz tilt toward $\hat{p}$, $\beta(0) = 0$) one has $\ell_\sigma \le a_2 P[\beta]$ for an explicit integral P, with equality along great circles through e_z and $\hat{p}$. For fixed $\beta^* = \beta(\sigma)$ the optimal profile is the lower envelope before σ and the upper envelope after, so $h_K = a_2 \max_{\beta^*} V(\beta^*)$ with $V = E + L_t$. Here E is nonincreasing (strictly on $[-T, 0]$, because $T < \pi/2 - 2\delta$), and L_t vanishes iff $\beta^* \le -\tau$, $\tau = T - \sigma$. Let $A(\sigma) = E(-\sigma)$ and $B(\sigma) = \max_{[-\tau,\sigma]} V$. At $\sigma = T$, $B < A$. At $\sigma = T/2 = h$, $A - V(h) \le -f(h)$ with $f(h) = \sin 2h - h\cos 2h + \sin h - 2h \ge h^3(\frac{1}{2} - \frac{2}{3}h^2) > 0$, so $B > A$. Hence there is a tie $\sigma_c \in (T/2, T)$. The two groups of maximizers are compact, disjoint, and separated by $\langle \hat{p}, I\rangle$: the left group coasts late, while the right group thrusts late and less early. A convex face cannot be a disjoint union of two nonempty compact sets, because convex sets are connected. □

D.2 Theorem E_0 (pure TVC, rest-terminal rate, tilt free)

Constant mass, so $\kappa = L/J_t = 1/(\varepsilon m)$ is constant. Roll rate zero is a model restriction. Write $f = T\cos\mu/m - \varepsilon\|\Omega_\perp\|^2$ for the axial drive of the centre of percussion y (Lemma 2.1). In the regime (H-f), (H-π), $f > 0$ and the tilt stays below $\pi/2$. Let $\mathcal{A}$ be the comparison class in which the transverse angular acceleration is bounded by $\kappa T \sin\mu$ with the gimbal deflection relaxed (its sign may chatter), and $\mathcal{P} \subseteq \mathcal{A}$ the planar meridional profiles. (L1) Rigidity: a body axis whose tilt equals the length of its path is a meridian. (L2) Comparison: $I_z(q)$ is bounded below by its value on the planar profile with angular speed $\|\Omega_q\|$, with equality iff the tilt equals the path length. (L3) Every minimizer of I_z over $\mathcal{P}$ tilts on a set of positive measure. (L4) I_z attains its infimum on $\mathcal{P}$ (weak-$*$ compactness of the throttle and Arzelà–Ascoli). Hence every $\mathcal{A}$-minimizer of I_z is meridional along one azimuth u with nonzero lateral increment. *Closure step.* The minimizer p^* uses chattering gimbal signs, so it need not be physical. On a grid of cells, use ordinary in-plane gimbal deflections $\pm\mu$ with a duty cycle chosen so that each cell's integrated lateral torque equals that of p^*. Then the body rate agrees with that of p^* at every node, including $\Omega_\perp(t_f) = 0$, and the axis stays planar. The axial thrust is unchanged, so $I(p_n) \to I(p^*)$, and the landing conditions on r, v are absorbed into the data. Thus $I(p^*)$ and $R_\pi I(p^*)$ lie in the velocity projection of $\operatorname{cl} F_{t_f}$. If $\operatorname{cl} F_{t_f}$ were convex, the midpoint $I_z(p^*)e_z$ would lie in it, and by compactness of the relaxed trajectory set it would be the increment of some $q \in \mathcal{A}$ minimizing I_z with zero lateral part, a contradiction. □

D.3 Theorem C-W (fixed attitude)

The extreme points of the cap $K_\Gamma(b)$ are exactly its spherical part $S_\Gamma(b) = \{\|T\| = \Gamma,\ \angle(T, b) \le \delta\}$, and $K_\Gamma = \operatorname{conv} S_\Gamma$ (Krein–Milman; Lean: `extremePoints_cap`, `cap_subset_closure_convexHull_sCap`). On each subinterval between prescribed nodes, the state increment is $\int \Psi_k(s) T(s)/m(s)\, ds$ with Ψ_k of full column rank and m fixed by Γ. Aumann's theorem applied to the multifunction $\Psi_k S_\Gamma(b)/m$ yields a measurable lossless selection with the same increment. Concatenation over subintervals preserves the state at every node. □

D.4 Theorem C-6 (pure TVC given attitude)

With zero roll rate the angular momentum is $J_\perp b \times \dot{b}$, and Euler's equation reads $J_\perp b \times \ddot{b} = L\,T \times b$. Splitting T and $\ddot{b}$ into components along and orthogonal to b gives $\Pi_b T = -k\Pi_b \ddot{b}$. So $T = T_a b + F$ with F determined by b. The gimbal condition becomes $\varkappa \le T_a \tan\delta$, and the throttle condition becomes $\rho_1^2 \le T_a^2 + \varkappa^2 \le \rho_2^2$. Both are intervals in T_a. The dynamics are affine in T_a for fixed b, and the fuel integrand $\|(T_a, \varkappa)\|$ is convex. □

E Proof of Theorem F

We prove Theorem 4.7. Notation is that of Section 4.5: $u = \ddot{y} - g$, $\dot{u} = y^{(3)}$, $\ddot{u} = y^{(4)}$, $\Pi_b = I - bb^\top$, $f = \|u\|$, and the flat map (6). The project note "10-landing-proof" in the supplementary archive gives the same proof with the test that checks each step numerically.

Lemma E.1 (The flat map is the solution)**.** *Let $y \in C^4([0,t_f];\mathbb{R}^3)$ with $u \neq 0$ everywhere and let $b, \dot{b}, a, \ell, m, T_a, T_\perp$ be given by (6). Under Assumption 4.6, (r, b, m) with $r = y - \varepsilon b$ and $\Omega = b \times \dot{b}$ solves the equations of model PT with zero roll rate,*

$$m\ddot{r} = T_a b + T_\perp + mg, \qquad J_t\dot{\Omega} = (-Lb) \times (T_a b + T_\perp), \qquad \dot{m} = -\alpha\|T_a b + T_\perp\|,$$

and it is the only solution with its initial values.

Proof. Differentiating $u = fb$ gives $\dot{u} = \dot{f}b + f\dot{b}$ and $\ddot{u} = \ddot{f}b + 2\dot{f}\dot{b} + f\ddot{b}$. Projecting, $\Pi_b\dot{u} = f\dot{b}$, so the $\dot{b}$ of (6) is the derivative of b, and

$$f\,\Pi_b\ddot{b} = \Pi_b\ddot{u} - 2\dot{f}\,\dot{b}, \qquad \dot{f} = \langle b, \dot{u}\rangle. \tag{9}$$

With $\Omega = b \times \dot{b}$ we have $\Omega \perp b$, $\Omega \times b = \dot{b}$ and $\dot{\Omega} = b \times \ddot{b} = b \times \Pi_b\ddot{b}$. *Rotation:* $(-Lb) \times (T_a b + T_\perp) = -Lb \times T_\perp = Lm\varepsilon\, b \times \Pi_b\ddot{b} = J_t\dot{\Omega}$, because $m\varepsilon L = J_t$. *Translation:* $m\ddot{r} = m(\ddot{y} - \varepsilon\dddot{b}) = m(fb + g - \varepsilon\ddot{b})$, and the thrust gives $T_a b + T_\perp + mg = m(f + \varepsilon\|\dot{b}\|^2)b - m\varepsilon(\ddot{b} + \|\dot{b}\|^2 b) + mg$, the same thing (we used $\Pi_b\ddot{b} = \ddot{b} + \|\dot{b}\|^2 b$). *Mass:* $\|T_a b + T_\perp\| = m\tau$ and $\dot{m} = -\alpha\tau m$ by the definition of m. The thrust history is continuous in t and the right-hand sides are locally Lipschitz in the state, so the solution is unique. □

Lemma E.2 (Boundary conditions)**.** *If y satisfies the boundary rows of (8) and rows (R1), (R2) at $t = 0$ and $t = t_f$, and $\angle(b_0, e_z) \le \vartheta_{\max}$, then $b(0) = b_0$, $\dot{b}(0) = \dot{b}_0$, $\Pi_b\ddot{b}(0) = \beta_0$, $r(0) = r_0$, $\dot{r}(0) = v_0$, and at t_f: $r = 0$, $\dot{r} = 0$, $b = e_z$, $\dot{b} = 0$, $T_\perp = 0$.*

Proof. End control points of a Bézier polygon are values of the curve, so (R2) gives $u_z \ge \ell_b > 0$ at both ends. At $t = 0$, $u(0) = f_0 b_0$ with $u_z(0) > 0$ and $(b_0)_z > 0$ forces $f_0 > 0$, hence $f(0) = f_0$ and $b(0) = b_0$. The second row is $\Pi_{b_0}\dot{u}(0) = f_0\dot{b}_0$, so $\dot{b}(0) = \dot{b}_0$ and $\dot{f}(0) = \langle b_0, \dot{u}(0)\rangle$. The third row and (9) give $\Pi_b\ddot{b}(0) = \beta_0$. Then $r(0) = y(0) - \varepsilon b_0 = r_0$ and $\dot{r}(0) = \dot{y}(0) - \varepsilon\dot{b}_0 = v_0$. At t_f, $u_h = 0$ and $u_z > 0$ give $b = e_z$; then $\dot{u}_h = 0$ gives $\dot{b} = 0$, and with $\dot{b} = 0$, $\ddot{u}_h = 0$ gives $\Pi_b\ddot{b} = 0$ by (9), i.e. $T_\perp = 0$. Finally $r = y - \varepsilon e_z = 0$ and $\dot{r} = \dot{y} - \varepsilon\dot{b} = 0$. □

Lemma E.3 (Pointwise bounds)**.** *Fix an instant and let $p = u_z > 0$. For all real η, κ, k, every $s \ge \sin\angle(b, e_z)$, every $0 < \ell_b \le p$ and every ϕ, put $x = \dot{u} - \eta u$ and let $q \ge \|x\|^2/p$ and $G \ge \|\ddot{u} - 2\eta\dot{u} - \kappa u - ke_z\| + s|k| + q$. Then*

$$\textit{(B1)}\ \ f\|\dot{b}\| \le \|x\|, \qquad \textit{(B2)}\ \ f\|\dot{b}\|^2 \le q, \qquad \textit{(B3)}\ \ f\|\Pi_b\ddot{b}\| \le G,$$
$$\textit{(B4)}\ \ p \le \tau \le \|(\|u\| + \varepsilon q/\ell_b,\ \varepsilon G/\ell_b)\|, \qquad \textit{(B5)}\ \ 2\phi p - \phi^2 \le f^2.$$

Proof. Note $p \le f$. (B1): $f\dot{b} = \Pi_b \dot{u} = \Pi_b x$ since $\Pi_b u = 0$. (B2): $f\|\dot{b}\|^2 = \|\Pi_b x\|^2/f \le \|x\|^2/p$. (B3): write $\dot{f} = \eta f + \langle b, x\rangle$ in (9). Then $2\dot{f}\dot{b} = 2\eta\Pi_b\dot{u} + 2\langle b, x\rangle \Pi_b x/f$ and

$$f\,\Pi_b\ddot{b} = \Pi_b(\ddot{u} - 2\eta\dot{u}) - 2\langle b, x\rangle\,\Pi_b x/f.$$

For the first term, $\Pi_b(\ddot{u} - 2\eta\dot{u}) = \Pi_b(\ddot{u} - 2\eta\dot{u} - \kappa u - ke_z) + k\Pi_b e_z$ and $\|\Pi_b e_z\| = \sin\angle(b, e_z) \le s$. For the second, $\langle b, x\rangle$ and $\Pi_b x$ are the components of x along and across b, so $2|\langle b, x\rangle|\|\Pi_b x\| \le \|x\|^2$, and the term is at most $\|x\|^2/f \le q$. (B4): $\tau \ge a \ge f \ge p$; and $a = f + \varepsilon\|\dot{b}\|^2 \le \|u\| + \varepsilon q/\ell_b$ by (B2), $\|\ell\| = \varepsilon\|\Pi_b\ddot{b}\| \le \varepsilon G/\ell_b$ by (B3), using $f \ge \ell_b$. (B5): $f^2 \ge p^2$ and $p^2 - (2\phi p - \phi^2) = (p - \phi)^2 \ge 0$. □

Lemma E.4 (Every instant, and the mass)**.** *Let y be feasible for (8). Then at every $t \in [0, t_f]$ there are numbers e_1, q, k, g_2, s_f and $\sigma(t) = \sum_j B_j(t)\sigma^j$ for which rows (R1)–(R6) hold with (U, U_1, U_2) replaced by $(u, \dot{u}, \ddot{u})(t)$. Moreover, with $z = \ln m$, on every span s*

$$z_{lo,s} \;\le\; z(t) \;\le\; z_{up,s}, \qquad \rho_1 \;\le\; m(t)\tau(t) \;\le\; \rho_2,$$

and $m(t_f) \ge m_0 \exp(-\alpha\frac{h}{4}\sum\sigma) \ge m_{dry}$.

Proof. Rows (R1)–(R6) define a convex set of points $(U, U_1, U_2, \text{auxiliaries})$ whose parameters are constant on a span. By (7), $((u, \dot{u}, \ddot{u})(t), \sum_j B_j(t)\,\text{aux}^j)$ is a convex combination of the four control tuples with their auxiliaries, so it lies in that set. By (R1) $\sin\angle(b, e_z) \le \sin\vartheta_{\max}$, so Lemma E.3 applies at every instant and gives $p(t) \le \tau(t) \le \sigma(t)$. Since $\dot{z} = -\alpha\tau$ and z decreases, $z(t) \le z(t_s) \le z_0 - \alpha\int_0^{t_s} u_z = z_{up,s}$ and $z(t) \ge z(t_{s+1}) \ge z_0 - \alpha\int_0^{t_{s+1}}\sigma = z_{lo,s}$, using $\int_0^{t_s} u_z = \dot{y}_z(t_s) - \dot{y}_z(0) + g_0 t_s$ and $\int_{\text{span}} B_j = h/4$. The function e^{-z} is decreasing and convex. A chord lies above it on the interval that (R7) confines $z_{lo,s}$ to, so $p(t) \ge \rho_1 c_s(z_{lo,s}) \ge \rho_1 e^{-z(t)}$, i.e. $m\tau \ge mp \ge \rho_1$. A tangent lies below it, so $\sigma(t) \le \rho_2\tau_s(z_{up,s}) \le \rho_2 e^{-z(t)}$, i.e. $m\tau \le m\sigma \le \rho_2$. The last claim is $z(t_f) \ge z_{lo,S}$ and the dry-mass row. □

Proof of Theorem 4.7. By (R2) and Lemma E.4, $u_z \ge \ell_b > 0$ at every instant, so the flat map is defined. Lemma E.1 gives the trajectory and its uniqueness. Lemma E.2 gives the initial and terminal conditions. At every instant, (R1) is the tilt bound, and

$$\varepsilon f\|\Pi_b\ddot{b}\| \;\le\; \varepsilon G \;\le\; \tan\delta\,(2\phi p - \phi^2) \;\le\; \tan\delta\, f^2$$

by (B3), (R5) and (B5), so $\|\ell\| \le \tan\delta\, f \le \tan\delta\, a$, which is the gimbal bound $\tan\mu \le \tan\delta$. The thrust bounds and the fuel bound are Lemma E.4. □

Proof of Proposition 4.8. The data (r_0, v_0) enter (8) only through the right-hand sides of its first two boundary rows, so the feasible set in the joint space of data and variables is convex and its projection onto (r_0, v_0) is convex. It is contained in the feasible boundary-data set by Theorem 4.7. A convex set cannot equal a nonconvex one. □

Table 2: Lean theorems corresponding to results in this paper. All compile without `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound` only.

Paper	Lean name (file)	Formal scope
Theorem 3.1, Proposition 2.5	`theoremA`, `not_isLossless_of_not_convex`, `not_isLosslessFor_of_not_convex`, `IsLosslessFor.isLossless` (`Abstract`)	any Θ, Y, any cost
Spherical triangle inequality and rigidity	`inner_triangle`, `inner_triangle_eq_lateral`, `lateral_eq_of_inner_ge` (`Sphere`)	inner-product form with equality case
Theorem 3.4 (variant F)	`not_convex_reachSet`, `not_convex_landingSet`, `no_lossless_relaxation_5dof`, `no_losslessFor_relaxation_5dof`, `landingSet_nonempty` (`Nonconvex`, `Main`)	$\delta + \omega t_f \le \pi/2$, any continuous $a_1 > 0$, any g, no state constraints, $b(t_f)$ free
Main Theorem; Theorem 3.7 (qualitative)	`no_lossless_relaxation_constrained_5dof`, `no_losslessFor_relaxation_constrained_5dof`, `not_convex_constrainedLandingSet`, `VStar_not_mem_WSet`, `witness_mem_glideSlope`, `exists_switchTime`, `constrainedLandingSet_nonempty` (`DiveBrake`, `Main`)	constant acceleration bounds, $g_0 < a_{\max}$, $0 < \delta < \pi/2$, every $t_f > 0$, every $\mathcal{X} \supseteq$ glide-slope set, $b(t_f)$ free
Main Theorem, upright (T′)	`not_convex_constrainedLandingSetUp`, `witness_admissible_upright`, `no_lossless_relaxation_constrained_upright_5dof` (`Upright`)	as above with $b(t_f) = e_z$; $2s^* \le t_f$ supplied for every t_f
Main Theorem, 6-DOF; Theorem 3.19	`Attitude6Admissible.toM1`, `lift_admissible`, `not_convex_landingSet6`, `not_convex_constrainedLandingSet6`, `not_convex_constrainedLandingSet6Up`, `no_lossless_relaxation_constrained_6dof`, `no_lossless_relaxation_constrained_upright_6dof` (`SixDOF`, `Upright`)	kinematic: proper rotations, total body rate $\le \omega$, free roll angle; free or upright terminal attitude
Main Theorem with mass depletion	`not_convex_constrainedLandingSet_mass`, `exists_max_phiFace_mass`, `no_lossless_relaxation_constrained_5dof_mass`, `no_losslessFor_relaxation_constrained_5dof_mass`, `no_lossless_relaxation_constrained_upright_5dof_mass`, `no_lossless_relaxation_constrained_6dof_mass`, `..._upright_6dof_mass` (`DiveBrakeMass*`)	`AccMass`: $0 < \rho_1 < \rho_2$, $\alpha > 0$, $0 < m_{dry} \le m_0 - \alpha\rho_2 t_f$, $0 \le g_0 < \rho_2/m_0$; every t_f
Theorem 3.7, free-time window	`no_lossless_relaxation_freeTime_5dof`, `no_lossless_relaxation_freeTime_5dof_mass` (`FreeTimeBStar`)	compact window; mass version needs $m_{dry} \le m_0 - \alpha\rho_2 t_b$
Theorem 3.7, quantitative	`landingSet_dist_ge`, `midpoint_not_mem_closure_landingSet`, `landingSet_dist_ge_mass` (`DiveBrakeQuant*`)	$\delta \le \pi/4$; max-norm, $\eta_0/(1+s^*)$; midpoint outside $\mathrm{cl}\, F$
Vehicle instance	`gtpl_no_lossless_upright_6dof_mass`, `gtpl_no_lossless_freeTime_mass` (`GTPLMass`)	400/1000 N, 79.88/55.88 kg, $I_{sp} = 180$ s, 15° cone, any ω, $t_f \le 42$ s; kinematic 6-DOF upright / 5-DOF free-time window
Lemma 2.1; pure-TVC skeleton (Theorem 3.22)	`TVC.cop_identity_E3`, `cop_identity_roll`, `convex_tvcSet_iff`, `tvc1_skeleton`, `tvc1_main`, `tvc1_model` (`TVC/`)	identity exact (constant ε); TVC theorem conditional on the certificate and analytic reduction as hypotheses
Theorem 3.4 with mass depletion	`AccMass`, `accMass_spec`, `no_lossless_relaxation_5dof_mass` (`MassModel`)	$0 < \rho_1 \le \rho_2$, $\alpha \ge 0$, $0 < m_{dry} \le m_0 - \alpha\rho_1 t_f$, $\delta + \omega t_f \le \pi/2$, no state constraints
Theorem 3.12 (abstract)	`StateLinearRelaxation.convexHull_subset`, `minError_lossy`, `convex_superset_gap` (`TheoremD`)	abstract; concrete instance not formalized
Positive side (namespace `LcvxPositive`, files `Positive/`; 35 audited theorems)		
Cap lemma (C-G, C-3, uniqueness, sharpness, extreme points)	`normal_of_max_of_norm_lt`, `norm_eq_or_antiRay_of_isMaxOn_cap`, `isMaxOn_cap_unique`, `extremePoints_cap`, `C3_headline` (`Cap`)	any real inner-product space; $\mathbb{R}^3$ headline

Table 3: Certified instances of Theorem 3.14 for a demonstration vehicle ($m_0 = 100$ kg, $\rho_1 = 1200$ N, $\rho_2 = 2500$ N, $\alpha = 0$).

δ	T [s]	Ω [rad]	margin M [m/s]	dent r_S [m]	witness offset $\lvert p_w \rvert$ [m]
0	2	0.40	1.39×10^{-2}	0.45	0.94
0	4	0.40	2.78×10^{-2}	1.80	3.74
5°	2	0.30	1.57×10^{-2}	0.48	1.79
10°	2	0.20	1.06×10^{-2}	0.39	2.59
10°	4	0.20	2.11×10^{-2}	1.57	10.38